Bulletin of Mathematical Biology (2026) 88:101
https://doi.org/10.1007/s11538-026-01663-x

Society for Mathematical Biology

ORIGINAL ARTICLE



Sustainable Coexistence with Infectious Diseases: A Behavioral Feedback Model Driven by Resource Accessibility under Static-Dynamic Optimal Control

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Received: 15 January 2026 / Accepted: 11 May 2026
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Abstract

This study develops a behavior-disease feedback model grounded in imitation dynamics to analyze optimal long-term epidemic coexistence strategies from both static and dynamic perspectives. Analytical findings reveal that the stability of the system depends on both the basic reproduction number and the behavioral threshold, with the latter and risk attention jointly determining optimal static intervention intensity. Data-driven analyses of influenza (Shanxi) and COVID-19 (Shanghai) show that adaptive public behavior can drive recurrent epidemic waves. In dynamic optimization simulations involving control of transmission rate alone or jointly with behavioral threshold, key insights include: seizing the critical rapid-growth window prevents both strategy oscillations and epidemic rebound resulting from delayed intervention; faster public response lowers both control costs and infection burden while accelerating attainment of control goals; stricter prevalence constraints require earlier and longer interventions, where proactive phased strategies outperform short, high-intensity measures in cost-effectiveness; achieving lower target prevalence necessitates earlier action and greater public caution, advancing the infection peak but substantially reducing its magni-

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Published online: 03 June 2026

tude. These results offer a theoretical framework and practical insights for optimizing resource allocation and guiding policy in endemic disease management.

Keywords Epidemic modeling · Behavioral adaptation · Healthcare resource accessibility · Optimal control · Endemic coexistence

1 Introduction

Traditionally, models of infectious disease transmission have often treated human populations as passive entities subject to external epidemiological shocks. However, real-world experiences from SARS to COVID-19 demonstrate that individuals actively perceive risks and adapt their behavior (Lau et al. 2004; Veltri et al. 2024). This creates a critical bidirectional interaction: epidemic dynamics drive protective decisions, while these adaptive behaviors, such as mask-wearing, subsequently influence disease spread, forming a complex feedback loop that shapes outbreak trajectories.

Critically, the public's perception of infection risk is grounded in observable realities. Medical resource accessibility, exemplified by metrics like hospital beds per capita, provides a tangible empirical basis for risk assessment (de Almeida et al. 2025; Huang and Chen 2024). When disease prevalence surpasses a psychological threshold linked to resource strain, concern intensifies, prompting proactive protection. Conversely, when prevalence recedes to perceived safe levels, reduced vigilance often leads to relaxed measures (Geoffard and Philipson 1996). This endogenous feedback mechanism, in which resource availability modulates behavior and in turn alters epidemic dynamics, remains insufficiently integrated into models that bridge theoretical insight and practical public health strategy.

Substantial research has aimed to formalize this interplay. Modeling approaches include integrating behavioral parameters into compartmental models (Capasso and Serio 1978; Perra et al. 2011) or adding compartments for behavioral states (Gozzi et al. 2025; Proverbio et al. 2025); utilizing multilayer and other complex networks (Granell et al. 2013, 2014; Nunner et al. 2021); employing agent-based models (Hoertel et al. 2020; Silva et al. 2020); and applying game theory (Bauch 2005; Li et al. 2024; Parino et al. 2024; Tang et al. 2022; Xiao et al. 2021; Zhao et al. 2018). The game-theoretic framework allows individuals to adapt strategies based on real-time payoffs, offering a powerful tool for modeling behavioral evolution during epidemics. Early self-learning models, however, emphasized rational choice while overlooking social learning (Chang et al. 2020). This gap is addressed by incorporating imitation dynamics, which capture how high-payoff strategies spread via an "observe, compare, and imitate" mechanism. This behavioral evolution affects disease transmission through two primary mechanisms: (1) altering contact patterns (e.g., self-isolation, changing commuting habits), which directly reduces individual contact frequency (Cai et al. 2025); (2) modifying infectiousness/susceptibility (e.g., mask-wearing, hand hygiene), which lowers the infection risk per contact (Parino et al. 2024; Zhao et al. 2019). In this integrated framework, individuals respond both to real-time epidemic data and to the observed choices of others, enabling a closed-loop feedback between behavioral evolution and disease transmission (Huang and Zhu 2022).

While these studies have significantly advanced the understanding of behavior-disease feedback mechanisms, their research has primarily focused on explaining phenomena such as multi-wave epidemic outbreaks and assessing the potential efficacy of voluntary vaccination strategies to achieve disease eradication (Bauch 2005; Bauch and Bhattacharyya 2012; Li et al. 2024; Xiao et al. 2021). A pivotal gap remains in the rigorous scientific formulation of optimal intervention strategies that explicitly integrate public health objectives (e.g., prevalence control) with socioeconomic costs, particularly given the prevailing paradigm of sustained long-term viral coexistence.

To address this gap, our study makes several key contributions. First, we develop a behavior-disease feedback model that explicitly positions healthcare resource accessibility as the central mediator of public risk perception, which then drives behavioral adaptation via imitation dynamics. Second, moving beyond explanatory modeling, we focus on designing and evaluating optimal control strategies for sustainable management in the context of long-term coexistence with the virus. Specifically, we introduce two control paradigms: (1) static optimization, which identifies a fixed transmission control level that minimizes the long-term cost of the associated endemic equilibrium (Seierstad and Sydsæter 1987); and (2) dynamic optimization, where time-varying strategies—applied either to the transmission rate alone or jointly to the transmission rate and the behavioral threshold—actively steer the system into a predefined neighborhood of the endemic equilibrium. This ensures that after interventions end, the system autonomously maintains a low prevalence level, thereby avoiding long-term reliance on intensive measures (Parino et al. 2024).

Finally, we apply this integrated framework to real-world data: influenza from Shanxi Province and COVID-19 from Shanghai. Our numerical analyses systematically investigate how factors such as optimization timing, behavioral response rates, prevalence constraints, and target prevalence levels influence the resulting optimal controls. The simulations yield evidence-based insights: seizing the early intervention window is critical to avoiding oscillatory control and epidemic resurgence; increasing public response rates produces a dual benefit by lowering both control costs and infection burden; under prevalence constraints, phased strategies offer greater cost-effectiveness compared to high-intensity centralized measures. Furthermore, achieving a lower and sustainable prevalence target requires both advancing the timing of interventions and strengthening public risk awareness, providing actionable guidance for long-term disease management.

2 Model Formulation

We propose an SIHS (Susceptible-Infected-Hospitalized-Susceptible) compartmental model that couples epidemic dynamics with adaptive human behavior. The population is stratified into three states: susceptible ($S(t)$), infected ($I(t)$), and hospitalized ($H(t)$). A key assumption is that recovery or hospital discharge confers no protective immunity, and individuals immediately become susceptible again. Therefore, $S(t) + I(t) + H(t) = 1$ for all $t \geqslant 0$, with $S(t), I(t), H(t) \in [0, 1]$.

2.1 SIHS Model

In the absence of behavioral adaptation, the natural progression of the epidemic is governed by the following system:

$$\begin{cases} I'(y) = \sigma\,(1 - I - H)\,I - \gamma_0 I - \delta_0 I, \\ H'(y) = \delta_0 I - \tau_0 H, \end{cases} \tag{1}$$

where σ represents the infection rate, γ_0 denotes the rate at which infected individuals recover and become susceptible again, δ_0 is the hospitalization rate of the infected, and τ_0 signifies the discharge rate. To reduce the number of parameters and simplify subsequent analysis, we rescale time by introducing $t = (\gamma_0 + \delta_0)y$. Under this transformation, the system (1) can be rewritten as the following form:

$$\begin{cases} I'(t) = R\,(1 - I - H)\,I - I, \\ H'(t) = \delta I - \tau H, \end{cases} \tag{2}$$

here, $R = \frac{\sigma}{\gamma_0+\delta_0}$ represents the reproduction number of the disease, $\delta = \frac{\delta_0}{\gamma_0+\delta_0}$ denotes the proportion of infectives being hospitalized, and $\tau = \frac{\tau_0}{\gamma_0+\delta_0}$.

2.2 Behavioral Evolution

To characterize the dynamic process of population behavior evolution, we model an individual's decision on whether to adopt self-protective behavior as a population game. In this game, individuals have two pure strategies: p (adopting protective measures) and u (remain unprotected). The payoffs for an individual's decision at time t are determined by the utility functions π_p and π_u corresponding to the chosen strategy (Chang et al. 2020).

The theory of "prevalence elastic behavior" proposes that individuals' willingness to adopt self-protective measures increases with the infection rate (Geoffard and Philipson 1996). Here, we assume that the perceived infection risk of individuals depends linearly on the prevalence I. Specifically, the perceived risks for individuals adopting and not adopting protective behaviors are characterized by the parameters ω_p and ω_u, respectively, with $\omega_p < \omega_u$, reflecting the reduction in perceived risk brought by protective measures. In addition, adopting protective behavior requires individuals to bear an inherent cost $k > 0$, such as the expenses associated with purchasing protective supplies. Under these assumptions, the utility functions corresponding to the two strategies are given by:

$$\pi_p(I) = -k - \omega_p I, \quad \pi_u(I) = -\omega_u I,$$

with the constraint $k < \omega_u - \omega_p$, since if $k \geq \omega_u - \omega_p$, it implies that not adopting protective behaviors always yields a higher benefit.

The simplest strategy updating rule is "imitate the better". Over time, individuals meet pairwise at random, and each individual has the opportunity to revise their own

strategy. If the randomly selected peer receives a higher payoff, the focal individual will switch to that superior strategy with a probability proportional to the payoff difference between them. This update rule is referred to as "imitation dynamics" (Chang et al. 2020; Sandholm 2010). The process can be captured by the payoff difference between the two strategies:

$$\alpha(I) = \pi_p(I) - \pi_u(I) = -k + (\omega_u - \omega_p)I,$$

which possesses the following properties: (1) $\alpha'(I) > 0$, reflecting the increased incentive for adopting self-protective behavior as the number of infections rises; (2) $\alpha(0) = -k < 0$, meaning that in the absence of infection, the fixed cost of protection makes forgoing such measures the cost-effective choice; (3) $\alpha(1) = -k + (\omega_u - \omega_p) > 0$, indicating that self-protective behavior becomes advantageous when the entire population is infected.

The behavioral evolution can be mathematically described by the imitation dynamics. Let $P(t) \in [0, 1]$ represent the proportion of individuals adopting protective strategies at time t. The behavioral evolution dynamics is governed by:

$$P'(t) = \varepsilon\left(\pi_p(I) - \pi_u(I)\right) P(1 - P) = \varepsilon\alpha(I) P(1 - P), \tag{3}$$

where $\varepsilon > 0$ is a rate capturing the speed of strategy imitation, which reflects individuals' sensitivity to payoff differences.

2.3 Coupled Epidemic and Behavioral Model

To reflect the bidirectional interplay between adaptive behavioral evolution and disease transmission, we establish a closed-loop feedback mechanism by modifying the infection rate (Parino et al. 2024). Specifically, we redefine R as a time-varying function that depends on the proportion of adopting self-protective behavior $P(t)$ in the population, i.e., $R(t) = R(P(t))$. The co-evolutionary coupled model is thus established as follows:

$$\begin{cases} I'(t) = R(P)(1 - I - H)I - I, \\ H'(t) = \delta I - \tau H, \\ P'(t) = \varepsilon\alpha(I) P(1 - P). \end{cases} \tag{4}$$

Furthermore, to quantify the population-level effect of protective behavior on disease transmission, we adopt a linear mixing formulation that links individual protective actions with the macroscopic infection level. Specifically, $R(P)$ is constructed as a weighted sum of the infection contributions from the unprotected and protected subpopulations. In many epidemiological scenarios, it is reasonable to assume that self-protective behaviors suppress transmission primarily by reducing infectivity. For instance, face masks reduce viral emission in respiratory infectious diseases, and condoms block the exchange of infectious body fluids in sexually transmitted diseases. Adopting such protective behaviors essentially lowers the per-contact transmission probability. To capture this effect, we introduce a parameter $a \in [0, 1]$ that quantifies the effectiveness of the protection, yielding

$$R(P) = R_0 \left[(1 - P) + (1 - a)\, P\right], \tag{5}$$

where R_0 represents the basic reproduction number of the disease. The formulation explicitly shows that $R(P)$ is a monotonic decreasing function of P, consistent with the epidemiological principle that increased protection reduces transmission risk. Substituting Eq. (5) into system (4) yields the following coupled behavior-disease model:

$$\begin{cases} I'(t) = R_0 \left[(1 - P) + (1 - a)\, P\right](1 - I - H)I - I, \\ H'(t) = \delta I - \tau H, \\ P'(t) = \tilde{\varepsilon} \left(\frac{I}{I_c} - 1\right) P\,(1 - P)\,. \end{cases} \tag{6}$$

Let

$$\Omega = \left\{(I, H, P) \middle| I \geq 0,\ H \geq 0,\ I + H \leq 1, 0 \leq P \leq 1\right\},$$

which is a positively invariant set of system (6). Consequently, system (6) is biologically well-posed and confines all subsequent analysis to this domain.

Here, $\tilde{\varepsilon} = k\varepsilon$ is a rate related to the speed of behavioral changes, and the decision threshold parameter $I_c = \frac{k}{\omega_u - \omega_p} \in (0, 1)$ is defined as the product of the number of beds per thousand people and a certain proportionality coefficient. I_c serves as a marginal early warning signal, directly linking the abstract risk perception parameters ω_u and ω_p to observable medical resource accessibility, thereby establishing a quantitative bridge between behavioral responses and resource constraints. This framework not only provides an analytical basis for understanding the triggering conditions of protective behaviors but also lays a methodological foundation for evaluating the regulatory effects of different medical resource allocation strategies on epidemic dynamics.

3 Existence and Stability of Equilibria

It follows from the third equation of system (6) that any equilibrium must satisfy one of the following conditions: $P = 0$, $P = 1$ or $I = I_c$. Accordingly, we analyze the equilibria in three cases. When $P = 0$, the corresponding equilibria are $DFE_1 = (0, 0, 0)$ and $EE_1 = \left(\frac{\tau}{\delta+\tau}\left(1 - \frac{1}{R_0}\right), \frac{\delta}{\delta+\tau}\left(1 - \frac{1}{R_0}\right), 0\right)$. In the case of $P = 1$, we obtain the equilibria $DFE_2 = (0, 0, 1)$ and $EE_2 = \left(\frac{\tau}{\delta+\tau}\left(1 - \frac{1}{(1-a)R_0}\right), \frac{\delta}{\delta+\tau}\left(1 - \frac{1}{(1-a)R_0}\right), 1\right)$. Finally, when $I = I_c$, the equilibrium is given by $EE_3 = \left(I_c, \frac{\delta}{\tau} I_c, P_5\right) = \left(I_c, \frac{\delta}{\tau} I_c, \frac{R_0(\tau - \tau I_c - \delta I_c) - \tau}{a R_0(\tau - \tau I_c - \delta I_c)}\right)$.

Among these, DFE_1 and DFE_2 are disease-free equilibria, while EE_1, EE_2 and EE_3 are endemic equilibria. Their existence and local stability are summarized in the following theorem.

Theorem 1 *The system (6) has at most five equilibria, and*

(1) DFE_1 *exists for all parameter regimes, and it is locally asymptotically stable if* $R_0 < 1$ *and an unstable saddle point if* $R_0 > 1$*;*

(2) EE_1 *exists iff* $R_0 > 1$*, and it is locally asymptotically stable iff* $I_c > \frac{\tau}{\delta+\tau}\left(1 - \frac{1}{R_0}\right)$*;*

Table 1 The existence and stability of all equilibria of system (6)

Equilibrium	Existence Conditions	Stability Conditions
DFE_1	–	$R_0 < 1$
EE_1	$R_0 > 1$	$I_c > \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right)$
DFE_2	–	×
EE_2	$(1-a)R_0 > 1$	$I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{(1-a)R_0}\right)$
EE_3	$R_0 > 1$ and one of the following conditions is satisfied: (a) $(1-a)R_0 > 1$ and $\frac{\tau}{\delta+\tau}\left(1-\frac{1}{(1-a)R_0}\right) < I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right)$; (b) $(1-a)R_0 \le 1$ and $I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right)$	√

[a] "–" indicates that the equilibrium exists in all parameter regimes;
[b] "×" indicates that the equilibrium is always unstable;
[c] "√" indicates that the equilibrium is stable, whenever it exists

(3) *DFE_2 exists for all parameter regimes and is always unstable;*

(4) *EE_2 exists iff $R_0 > \frac{1}{1-a}$, and it is locally asymptotically stable iff $I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{(1-a)R_0}\right)$;*

(5) *EE_3 exists and is locally asymptotically stable iff $\tau - \tau I_c - \delta I_c > 0$ and one of the following conditions holds: (a) $1 < R_0 \le \frac{1}{1-a}$ and $I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{(1-a)R_0}\right)$, or (b) $R_0 > \frac{1}{1-a}$ and $\frac{\tau}{\delta+\tau}\left(1-\frac{1}{(1-a)R_0}\right) < I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right)$.*

The complete proof of this theorem is provided in Appendix A, established through rigorous analysis of the Jacobian matrix and its eigenvalues. The corresponding conditions are summarized in Table 1.

Furthermore, the stability regions of all equilibria are synthesized in a two-parameter bifurcation diagram in the $(R_0,\ I_c)$ plane (Fig. 1). This diagram reveals that, when $R_0 > 1$, I_c determines the final state of the system: (1) A reduction in healthcare accessibility can shift the equilibrium from EE_1 to EE_3, resulting in fewer infected individuals while more people adopt self-protective behaviors. (2) If $\frac{1}{1-a} < R_0$, a further reduction in healthcare accessibility can shift the equilibrium from EE_3 to EE_2, further reducing infections and leading to universal adoption of protective measures. These results indicate that moderate strain on medical resources may stimulate proactive protective behaviors among the public, ultimately reducing disease prevalence. This challenges the conventional view that more resources mean greater safety and highlights the vital role of the behavioral threshold as a risk signal. The key insight is that more efficient epidemic control can be achieved by adopting threshold-based management to activate public self-adaptation, rather than by simply increasing resources.

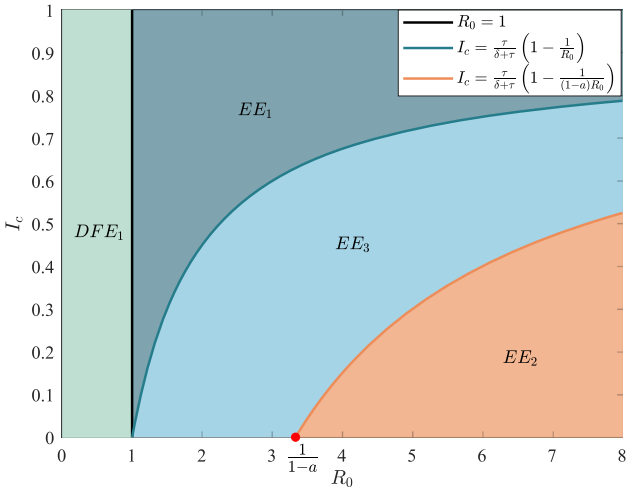


Fig. 1 Bifurcation diagram in the (R_0, I_c) parameter plane for system (6). The stability regions of the equilibria are indicated, with labels denoting the corresponding asymptotic states. Parameter values are $a = 0.7$, $\tau = 0.09$ and $\delta = 0.01$ (color figure online)

4 Static and Dynamic Intervention Policies under Single Control

Non-pharmaceutical interventions (NPIs) are central to epidemic control, with their intensity and timing directly influencing both containment effectiveness and socioeconomic costs. To quantify those effects, we introduce a transmission reduction factor $(1 - u)$. Specifically, we assume that NPIs (e.g., social distancing, mask mandates) can reduce the basic reproduction number R_0, which quantifies transmission in the absence of protective conditions, by a factor of $(1-u)$. This is captured by the parameter $R_1 = (1 - u)R_0$, where the control variable $u \in [0, 1]$ denotes intervention intensity, $u = 0$ corresponds to no intervention, and $u = 1$ indicates the complete interruption of transmission.

4.1 Static Intervention Policies

Based on the above setup, we begin by analyzing the scenario of constant intervention intensity and establish the corresponding controlled dynamic system:

$$\begin{cases} I'(t) = (1-u)\, R_0\, [(1-P) + (1-a)\, P]\, (1-I-H)I - I, \\ H'(t) = \delta I - \tau H, \\ P'(t) = \tilde{\varepsilon}\left(\frac{I}{I_c} - 1\right) P\,(1-P)\,. \end{cases} \tag{7}$$

To formalize the optimal control problem (OCP), we define a cost functional that captures the trade-off between disease prevalence and intervention cost:

$$J(u) = hI(u) + (1-h)u^2, \tag{8}$$

where the two terms are weighted by the parameter $h \in [0, 1]$, a high h prioritizes reducing infections, while a low h emphasizes minimizing the economic costs of interventions.

In static control, our objective is to determine the optimal fixed intervention strategy u^* and the corresponding equilibrium $x^* = (I^*, H^*, P^*)$ that jointly minimize the cost functional (8), i.e., $u^* = \arg\min_{u\in[0,1]} J(u)$. Theorem 1 shows that the stability of the DFE depends on the control input u. When $(1-u)\,R_0 < 1$, the $DFE_1 = (0, 0, 0)$ is stable, and the cost functional depends solely on u. The optimal control in this case is $u^* = \max\{0, (1 - 1/R_0)^+\}$, indicating that no control is required if $R_0 < 1$, while partial control is implemented for $R_0 > 1$ to maintain the disease-free state. Conversely, if $(1-u)\,R_0 > 1$, the system (7) converges to an endemic equilibrium (EE). Although multiple candidate equilibria EE_1, EE_2 and EE_3 may exist, stability analysis confirms the uniqueness of the stable EE, which is determined by I_c. Thus, we focus on this scenario, where the EE is a function of the control input, which is denoted as $x^*(u)$ and obtained by replacing R_0 with $(1-u)R_0$ in the expressions from Section 3. We evaluate the cost functional J at EE, which corresponds to one of EE_1, EE_2 or EE_3. By varying u, the cost functional at this stable equilibrium forms a piecewise continuous function composed of three segments, i.e.,

$$J(u) = \begin{cases} J_1(u) := h\frac{\tau[(1-u)R_0-1]}{(\delta+\tau)(1-u)R_0} + (1-h)u^2 & if\ u \in \Omega_1, \\ J_2(u) := h\frac{\tau[(1-u)(1-a)R_0-1]}{(\delta+\tau)(1-u)(1-a)R_0} + (1-h)u^2 & if\ u \in \Omega_2, \\ J_3(u) := hI_c + (1-h)u^2 & if\ u \in \Omega_3. \end{cases} \tag{9}$$

Here, $\Omega_1 = (1 - C, u_u]$, $\Omega_2 = [0, 1 - D)$, $\Omega_3 = [1 - D, 1 - C]$, where $C = \frac{\tau}{R_0(\tau-\tau I_c-\delta I_c)}$, $D = \frac{\tau}{(1-a)R_0(\tau-\tau I_c-\delta I_c)}$, and $u_u = 1 - \frac{1}{R_0} - \xi$, with $\xi > 0$ sufficiently small. To determine the optimal control input, we focus on analyzing the monotonicity of the cost functional for $u \in [0, 1]$ and further compare its local minima across the three intervals, as summarized in the following theorem.

Theorem 2 *For system (7) with the cost functional (8), the optimal control strategy can only be selected from the finite candidate set* $\{0, v_1, u_1, 1 - D, u_u\}$.

The detailed proof and the definitions of parameters A, B, u_1, u_2, v_1 and v_2 are provided in Appendix B. Building on this theorem, the global minimum of the cost functional for all cases is summarized in Table 2. Although the complexity of the cost functional precludes an analytical characterization of the full parameter regions corresponding to each optimal control, for any given parameter configuration, the optimal strategy can be identified by comparing only the relevant subset of candidate solutions, determined by the monotonicity and stationary points of each segment. This derived control, in combination with Theorem 1, uniquely determines the corresponding endemic equilibrium. We further employ numerical investigation to determine the optimal inputs and the associated endemic equilibria under varying parameter conditions. Simulations for the cost functional across different scenarios shown in Figs. 2, 3, 4.

Table 2 Global minimum of the cost functional under different cases

Conditions		Global Minimum of the Cost Functional
$1-C \le 0$	(i)	$J(u_u)$
	(ii) or (iii), $u_1 > u_u$	
	(ii) or (iii), $u_2 < u_u$	$\min\{J(u_1), J(u_u)\}$
	(ii) or (iii), $u_1 < u_u < u_2$	$J(u_1)$
$1-D \le 0 < 1-C$	(i)	$\min\{J(0), J(u_u)\}$
	(ii) or (iii), $u_2 < 1-C$ or $u_1 > u_u$	
	(ii) or (iii), $u_1 < 1-C < u_2 < u_u$	
	(ii) or (iii), $u_1 < 1-C$ and $u_u < u_2$	$J(0)$
	(ii) or (iii), $1-C < u_1 < u_2 < u_u$	$\min\{J(0), J(u_1), J(u_u)\}$
	(ii) or (iii), $1-C < u_1 < u_u < u_2$	$\min\{J(0), J(u_1)\}$
$0 < 1-D$	(i)	$\min\{J(1-D), J(u_u)\}$
	(ii), $u_2 < 1-C$ or $u_1 > u_u$	
	(ii), $u_1 < 1-C < u_2 < u_u$	
	(ii), $u_1 < 1-C$ and $u_2 > u_u$	$J(1-D)$
	(iii), $v_1 > 1-D$, $u_1 < 1-C$ and $u_2 > u_u$	
	(ii) or (iii), $1-C < u_1 < u_2 < u_u$	$\min\{J(1-D), J(u_1), J(u_u)\}$
	(ii) or (iii), $1-C < u_1 < u_u < u_2$	$\min\{J(1-D), J(u_1)\}$
	(iii), $v_2 < 1-D$ and $u_2 < 1-C$	$\min\{J(v_1), J(1-D), J(u_u)\}$
	(iii), $v_2 < 1-D$ and $1-C < u_2 < u_u$	
	(iii), $v_2 < 1-D$ and $u_2 > u_u$	$\min\{J(v_1), J(1-D)\}$
	(iii), $v_1 < 1-D < v_2$ and $u_2 < 1-C$	$\min\{J(v_1), J(u_u)\}$
	(iii), $v_1 < 1-D < v_2$ and $1-C < u_2 < u_u$	
	(iii), $v_1 < 1-D < v_2$ and $u_2 > u_u$	$J(v_1)$
	(iii), $u_1 > u_u$ or $1-D < v_1 < u_2 < 1-C$	$\min\{J(1-D), J(u_u)\}$
	(iii), $v_1 > 1-D$ and $u_1 < 1-C < u_2 < u_u$	

[a] The labels (i)-(iii) stand for the following mutually exclusive parameter regimes: (i) $B > A \ge \frac{4}{27}$, (ii) $B \ge \frac{4}{27} > A$, (iii) $\frac{4}{27} > B > A$. The parameters A, B, u_1, u_2, v_1 and v_2 are defined in Appendix B

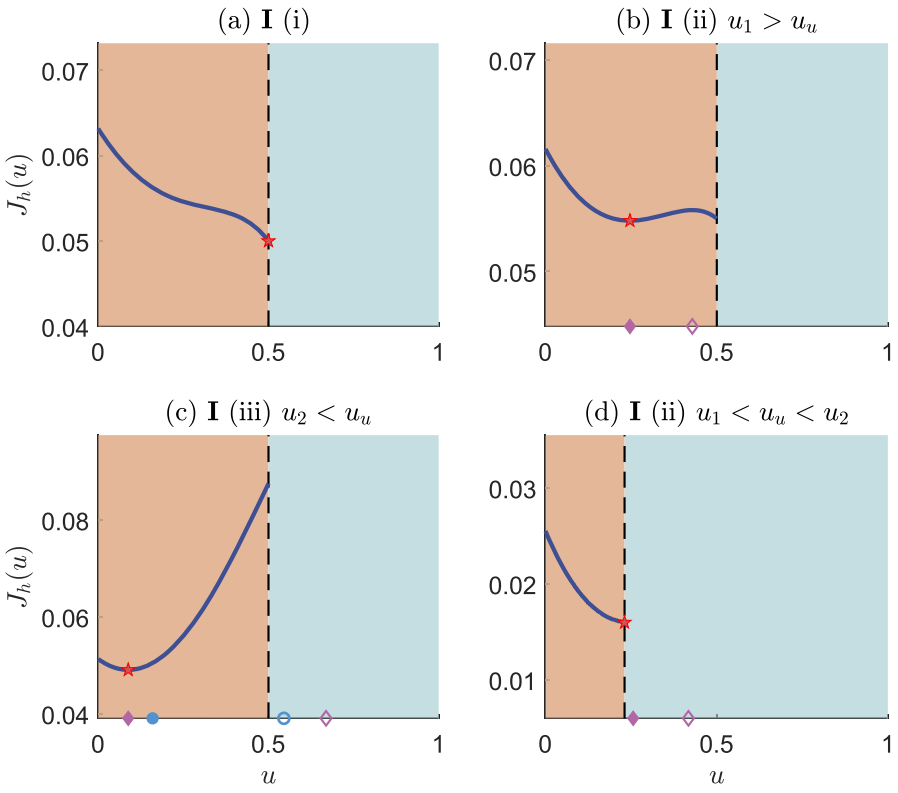


Fig. 2 Simulations of the cost functional under **I** (i.e. $1 - C \leq 0$). Solid diamonds, hollow diamonds, solid circles, hollow circles, and pentagrams identify u_1, u_2, v_1, v_2, and the optimal input u^*, respectively. The dashed, dash-dotted and dotted lines represent the regional boundaries u_u, $1 - C$ and $1 - D$, respectively. The labels (i)-(iii) stand for the following mutually exclusive parameter regimes: (i) $B > A \geq \frac{4}{27}$, (ii) $B \geq \frac{4}{27} > A$, (iii) $\frac{4}{27} > B > A$ (color figure online)

Remark 1 It should be noted that under the static control strategy, the cost functional implicitly accounts for hospitalization through a linear relationship. At equilibria, hospitalizations are proportional to infections $H = \delta I/\tau$. Thus, by adjusting the weighting parameter h, hospitalization costs can be indirectly reflected in the infection term hI, without modifying the cost functional's structure. This approach maintains operational simplicity while providing policymakers with the flexibility to balance transmission risk and economic costs by tuning the parameter h.

Remark 2 Analysis shows that the optimal control input depends critically on the model parameters a and I_c, as well as the cost trade-off coefficient h. Specifically, I_c defines the boundary $1 - C$, and together with a, jointly determines the boundary $1 - D$. These boundaries partition the control space and determine the specific interval where $u = 0$ belongs. Meanwhile, h governs the magnitude of A, and together with a, regulates B. Through their influence on the existence and regional affiliation of u_1, u_2, v_1 and v_2, A and B respectively determine the monotonicity of the cost functional over

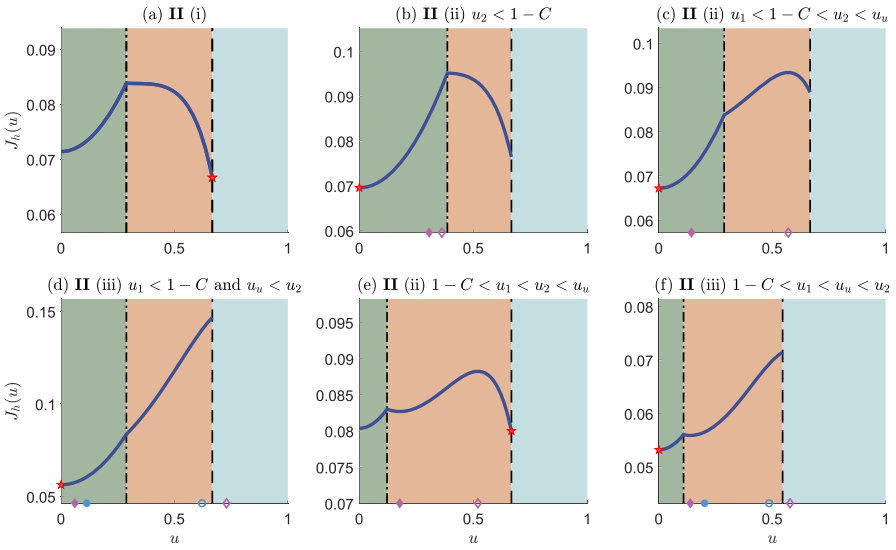


Fig. 3 Simulations of the cost functional under **II** (i.e. $1 - D \leq 0 < 1 - C$). The markers and line styles follow the same convention as in Fig. 2 (color figure online)

intervals Ω_1 and Ω_2. Changes in these three key parameters (I_c, a, and h) may directly alter the curve shape and values of the cost functional, subsequently influencing the optimal control inputs. This phenomenon can be directly observed in **I**(iii): $\frac{4}{27} > B > A$, as shown in Fig. 4.

Then, we demonstrate how parameters h and I_c affect the optimal control input u^* and the convergence dynamics of the system (7) in Fig. 5, revealing several meaningful phenomena.

At low values of h ($h < 0.6$), three scenarios emerge. When $I_c < 0.14$, the low behavioral threshold causes even a small number of infections to trigger widespread adoption of protective measures among individuals, without the need for strong external control. System (7) converges to EE_2, leading to a stable equilibrium characterized by universal adoption of self-protective behavior. When $I_c \in (0.145, 0.17)$, $u^* = 0$ (no additional control is applied), and the system converges to EE_3, corresponding to partial self-protection within the population. When $I_c > 0.18$, the high behavioral threshold weakens the population's perception of infection risk, thereby inhibiting the spontaneous adoption of protective measures. To reduce associated costs, an external control ($u^* \neq 0$) is necessary to drive the system to EE_1.

At sufficiently large values of h (red region), the cost of infection risk becomes the dominant factor. Regardless of the level of I_c, strong control must be implemented to effectively suppress infections, which leads to the suppression of public self-protection and drives the system to EE_1, forming an inverse relationship between control intensity and protection demand.

At intermediate values of h (colored region), the behavioral threshold I_c affects both the optimal control strategy and the final steady state. Specifically, when $I_c < 0.18$, the system (7) converges to EE_2 (full protection) or EE_3 (partial protection), and

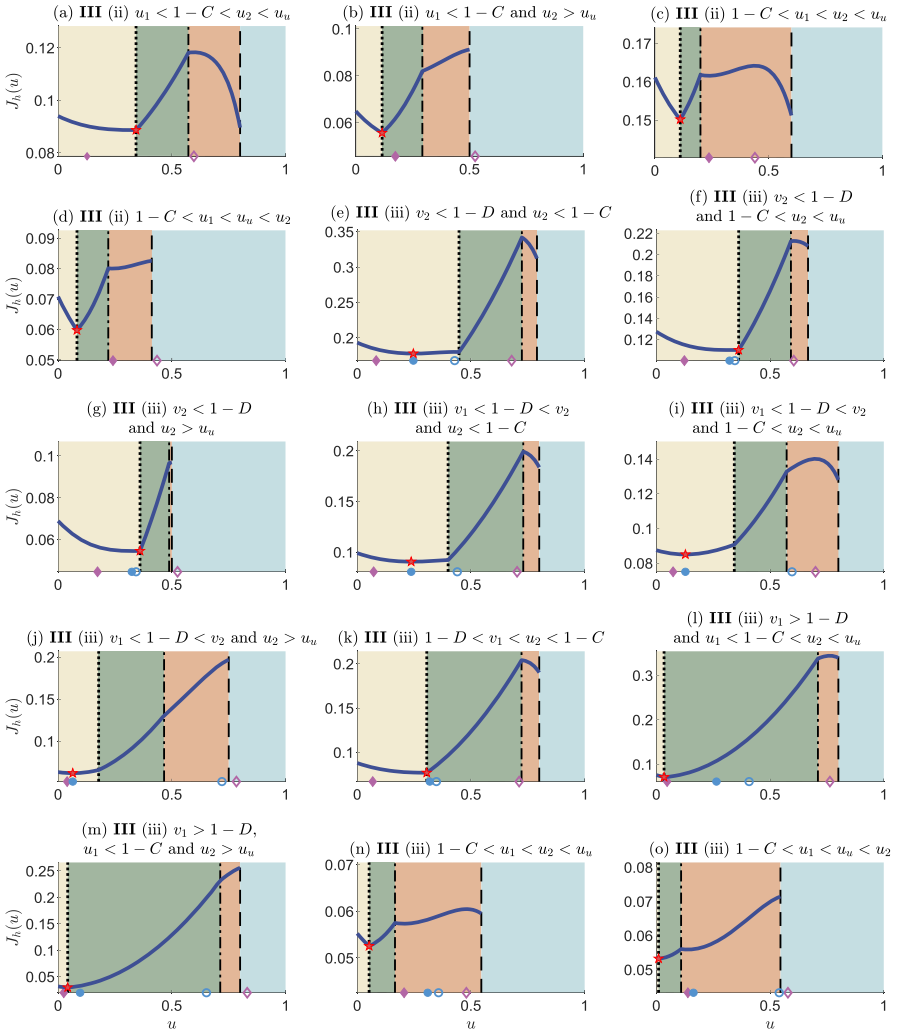


Fig. 4 Simulations of the cost functional under **III** (i.e. $0 < 1 - D$). The markers and line styles follow the same convention as in Fig. 2 (color figure online)

minor changes in I_c can cause switches between them, indicating that an appropriate selection of the risk signal is crucial for system (7) to stabilize at the desired steady state. Interestingly, although an increase in I_c weakens the population's perception of disease risk, it reduces the optimal control input u^*. This finding reveals a complex trade-off between individual behavioral decisions and macro-level disease control strategies.

Furthermore, as h increases, u^* shows a non-decreasing trend, indicating that when decision-makers place greater emphasis on the risk costs related to infection numbers, stronger implementation of control measures is required to meet stricter requirements for infection reduction. It is worth noting that u^* will experience jump-like growth in

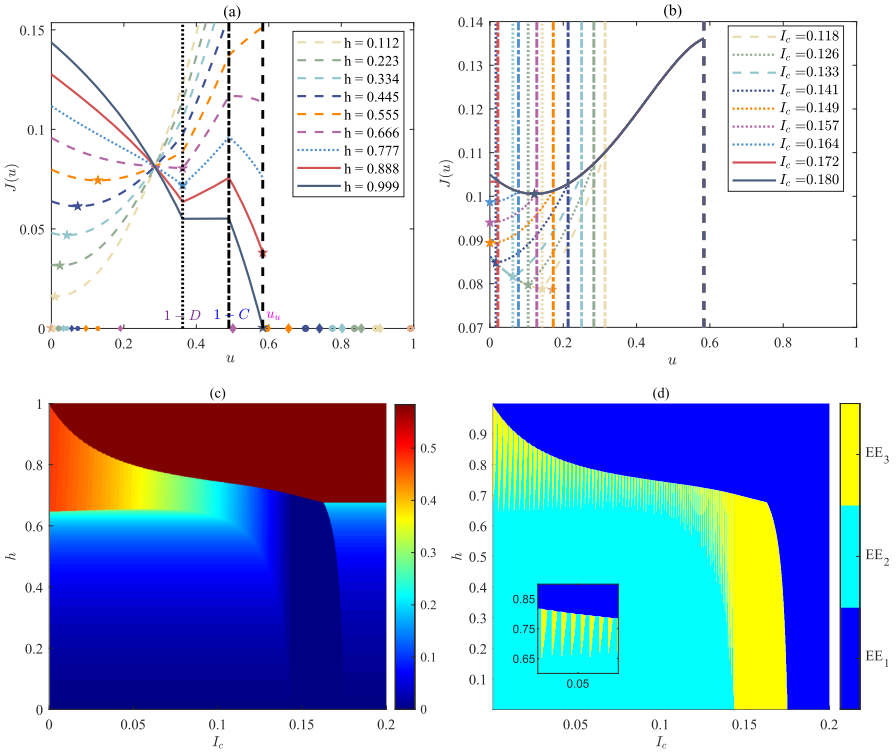


Fig. 5 Optimal control strategies and system convergence dynamics under different medical insurance impact weight (h) and behavioral decision-making threshold (I_c). The parameters are set as $R_0 = 2.4$, $\delta = 0.7$, $\tau = 0.3$ and $a = 0.2$. (a) Results under different h with $I_c = 0.055$. (b) Results under different I_c with $h = 0.6$. (c) Optimal strategies under different h and I_c. (d) Convergence dynamics of the system for the parameter region depicted in panel (c). Solid diamonds, hollow diamonds, solid circles, hollow circles, and pentagrams denote u_1, u_2, v_1, v_2, and the optimal input u^*, respectively. The solid, dashed, and dotted lines represent the system converging to EE_1, EE_2 and EE_3 under the optimal control. The vertical dashed, dash-dot and dotted lines represent the corresponding demarcation boundaries u_u, $1 - C$ and $1 - D$, respectively (color figure online)

a specific interval, reflecting a qualitative shift in disease prevention strategies: when the emphasis on infection risk exceeds a critical threshold, the prevention and control modes switch abruptly from relatively moderate to high-intensity interventions.

4.2 Dynamic Intervention Policies

The previous section focused on determining an optimal fixed intervention strategy and its corresponding equilibrium. We now turn to dynamic optimization: designing a time-varying control $u(t)$ over a finite horizon T to drive the system from an given initial state $x_0 \neq x^*$ to the target equilibrium x^* with minimal cumulative cost. To achieve this objective, we formulate a cost functional consisting of two components: the integral of the instantaneous cost along the system trajectory and a terminal penalty for deviation from the desired state at the terminal time. Specifically, in the time-

varying context, the instantaneous cost is defined by extending the static framework in Eq. (8) as follows

$$J(x(t), u(t)) = hI(t) + (1-h)u^2(t),$$

and the terminal penalty is defined as

$$\Phi(x(T)) = \psi \|x(T) - x^*\|_2,$$

where $\psi \geq 0$ denotes the terminal cost weighting coefficient whose value quantifies the system's tolerance for final state deviation. This design, while not guaranteeing the system reaches the target state precisely at time T, avoids the infeasibility of optimization problems caused by rigid state constraints, ensures the system trajectory remains close to the target equilibrium, and provides an implementable decision-making framework for dynamic resource optimization in complex epidemic scenarios.

The state evolution is governed by the following non-autonomous nonlinear system:

$$\begin{cases} I'(t) = (1-u(t))\, R_0\, [(1-P) + (1-a)\, P]\, (1-I-H)I - I, \\ H'(t) = \delta I - \tau H, \\ P'(t) = \tilde{\varepsilon} \left(\frac{I}{I_c} - 1 \right) P\, (1-P)\,. \end{cases} \tag{10}$$

Furthermore, in practical epidemic control, completely interrupting disease transmission (i.e., $u = 1$) is often unattainable due to objective factors such as population compliance limitations and the necessity to maintain basic socioeconomic activities. To address this, we introduce a practical upper bound $u_{\max} \in (0, 1)$ such that $u(t) \in [0, u_{\max}]$ holds for all t. Besides, we define the admissible control set $U = \{u \mid u(t)\ is\ measurable\ on\ [t_0, T],\ 0 \leq u(t) \leq u_{\max},\ \forall\, t \in [t_0, T]\}$.

Integrating the above components, the dynamic OCP is formally stated as:

$$\begin{aligned} &\min_{u(t)} \int_{t_0}^{T} J(x(t), u(t))\, dt + \Phi(x(T)), \\ \text{subject to} \quad &\text{Eq. (10)}, \\ &x(t_0) = x_0, \\ &0 \leq u(t) \leq u_{\max}. \end{aligned} \tag{11}$$

Applying Theorem 4.1 of (Fleming and Rishel 1975) ensures the existence of a solution to the OCP (11), as stated in the following theorem.

Theorem 3 *For the OCP (11) with initial conditions* $(I_0, H_0, P_0) \in \Omega$*, there exists an optimal control* $u^*(t)$ *that minimizes the cost functional over the admissible control set* U.

The characterization of the optimal control is provided by Pontryagin's Maximum Principle (PMP) (Pontryagin 1987), as stated in the following theorem.

Theorem 4 *For the OCP (11), there exist adjoint variables* $\boldsymbol{\lambda}(t) = [\lambda_1(t), \lambda_2(t), \lambda_3(t)]^\top$, *and an optimal control* $u^*(t)$, *which minimizes the cost functional over the admissible control set* $U = [0, u_{\max}]$, *and* $u^*(t)$ *satisfies the following necessary condition:*

$$u^*(t) = \min\left\{u_{\max}, \max\left\{0, \frac{\lambda_1(t)R_0(1-aP(t))(1-I(t)-H(t))I(t)}{2(1-h)}\right\}\right\},$$

and the adjoint variables satisfy the transversality condition associated with the terminal cost $\boldsymbol{\lambda}(T) = \nabla_x \phi(x(T))$.

The proof of Theorem 3 is provided in Appendix C. Detailed derivations of the adjoint functions λ_i $(i = 1, 2, 3)$ and the solution of the OCP (11) are given in Appendix D.

4.3 Case Study: Application to Influenza

We use influenza as a case study to demonstrate the feasibility of the proposed method. Influenza is an acute respiratory disease to which the general population is generally susceptible (Javanian et al. 2021), and is strongly influenced by human behavior. Although most cases are mild and self limited, it still poses a non-negligible threat to global health and economies. According to World Health Organization (WHO) statistics, influenza leads to approximately 3 to 5 million hospitalizations worldwide each year (WHO 2025). In the United States alone, it leads to 3.7 million outpatient visits, 650,000 emergency department visits, and 247,000 hospitalizations annually, with associated economic costs reaching tens of billions of dollars (Putri et al. 2018). These data highlight the necessity of effective prevention strategies to reduce transmission and mitigate the societal burden of influenza.

4.3.1 Model Parameters

The parameter values for this model were informed by epidemiological data specific to Shanxi Province, influenza characteristics documented in the North China region, local clinical observations, and findings from established literature.

Total population N: fixed at 36.5612 million (average year-end population of Shanxi, 2012-2017) (Shanxi Provincial Bureau of Statistics 2019). Natural recovery rate: $\gamma = 1/10$ (with an average recovery period of 10 days, which is the midpoint of the typical 1-2 week influenza course). Hospital discharge rate: $\tau = 1/8$, based on influenza research conducted in Beijing (also located in North China) during 2014-2015 (Zhang et al. 2015). Initial hospitalized cases: $H_0 = 145/N$, based on the number of newly confirmed cases from December 24-31, 2011. Given their widespread use in respiratory infection control, masks are incorporated into our model with a risk reduction parameter of $a = 0.45$, based on findings that it reduces influenza infection risk by approximately $45\%(95\%CI[0.39 - 0.76])$ (Liang et al. 2020).

We validated the proposed model against reported influenza cases from Shanxi Province (2012-2017). The simulated new hospitalizations show a good fit with

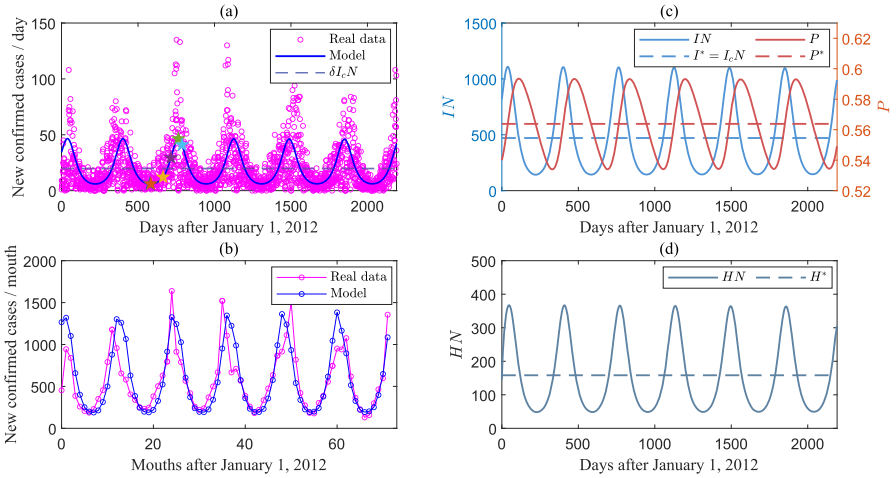


Fig. 6 Model fitting to influenza data from Shanxi Province, China. Panels (a) and (b) show the daily and monthly numbers of newly confirmed cases, respectively, and the corresponding model-fitted curves. Key model parameters are estimated: $R_0 = 1.34$, $\tilde{\varepsilon} = 2.2 \times 10^{-3}$, $I_c = 1.288 \times 10^{-5}$, $\delta = 0.042$, $I_0 = 2.237 \times 10^{-5}$ and $P_0 = 0.54$. (c) Simulated numbers of infections and the proportion of the population adopting protective behavior. (d) Simulated numbers of hospitalized cases (color figure online)

surveillance data (Fig. 6(a)). A clear lag exists between epidemic peaks and the peak in protective behavior. This lag arises because rising infections act as a risk signal, triggering protective measures which in turn suppress transmission. However, protective behavior exhibits inertia, with public caution leading to its persistence until infections remain low, thereby causing the observed peak offset.

The 6% variation in protective behavior suggests a limited public response, consistent with the established public perception of influenza as a self-limited and lower-risk illness. This level of behavioral response can mitigate but not halt disease transmission, creating a persistent tug-of-war between the epidemic and protective behavior. This endogenous feedback mechanism itself drives the recurrent, multi-peak outbreaks. Thus, breaking this cycle necessitates targeted external interventions.

It should be noted that the above multi-wave oscillations are not genuine periodic solutions. Under the estimated parameters, the system possesses a unique stable focus. Due to the small absolute value of the negative real part of the eigenvalues, the convergence rate is slow. Consequently, the multi-wave oscillations presented in Fig. 6(a,b) represent damped oscillations with slowly decaying amplitudes. Extending the simulation time would reveal that the system ultimately converges to this stable equilibrium.

4.3.2 Optimal Intervention

Simulations based on influenza data from Shanxi province show persistent, multi-wave outbreaks (Fig. 6), indicating that complete eradication remains challenging. To address the practical need for rapid epidemic control, we formulate a minimum-time control problem, which aims to dynamically adjust interventions over the shortest

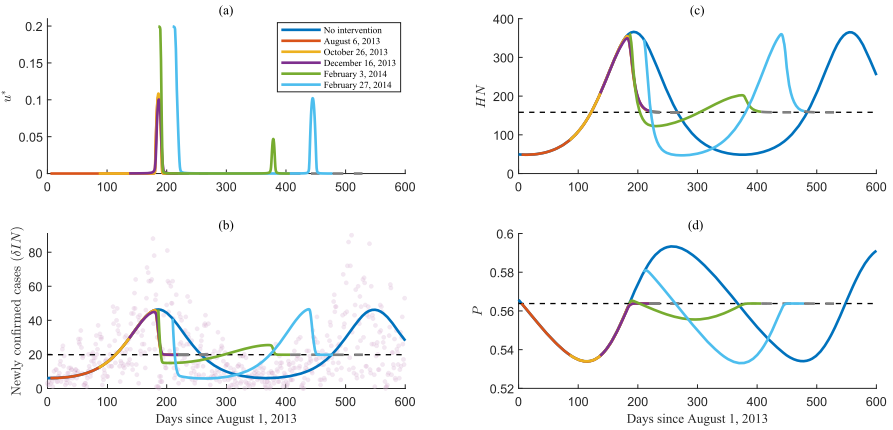


Fig. 7 Results of the OCP (11) under different optimization starting times. Parameter values are $h = 0.6$, $\psi = 3 \times 10^6$. (a) Optimal control curve u^*. (b-d) Time series of the number of newly confirmed cases, hospitalized individuals, and the proportion of the population adopting protective behavior under optimal control, respectively. The black dashed line in each subplot indicates the equilibrium values of the corresponding variables for the uncontrolled system (6), and the pink scatter points represent real data (color figure online)

possible time T such that the system enters a predefined neighborhood (10^{-7}) of the endemic equilibrium x^*. The minimal time T is numerically identified by incrementally extending the horizon and solving the control problem until the terminal condition is satisfied. Importantly, this terminal condition serves as a stopping criterion for the minimum-time search rather than a hard constraint, with its precision threshold adjustable according to practical objectives. Unlike free terminal time optimal control, here T is not a decision variable, and there is no need to compare total costs over different time horizons. This ensures that post-intervention, the system avoids significant deviation from x^*, preventing new epidemic waves and maintaining sustainable, low-level endemic coexistence.

The time-varying control variable $u(t)$ is bounded by $[0, 0.2]$, which is consistent with influenza's self-limiting nature and empirical evidence that measures such as restricting large gatherings, closing venues, and suspending classes reduce mobility by about $20\% - 25\%$. This upper bound also minimizes disruption to daily life. By solving the OCP, we obtain the optimal control strategy for each scenario, which avoids both insufficient control leading to outbreak escalation and excessive intervention causing unnecessary societal and economic disruptions. This enables manageable steady-state control of influenza.

First, using the parameters estimated in Section Fig. 6 as the baseline, we simulate how the optimization starting time affects the optimal control and intervention dynamics. The results, presented in Fig. 7, demonstrate that the optimization starting time critically determines both the attainment of an acceptable endemic equilibrium and the structure of the optimal control. During the epidemic's growth phase, a critical window exists where optimization initiation yields a structurally stable optimal strategy, with nearly identical control trajectories and state paths across varied initial times.

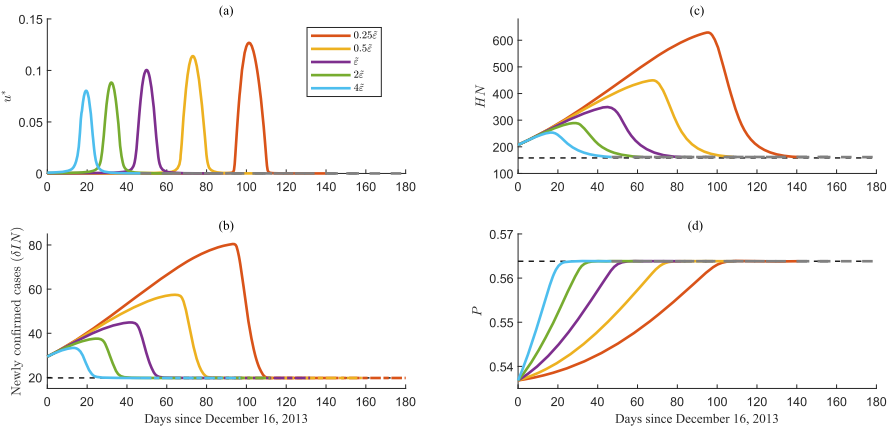


Fig. 8 Results of the OCP (11) under different behavioral response rates $\tilde{\varepsilon}$. The initial values are $I_0 = 1.9153 \times 10^{-5}$, $H_0 = 5.6638 \times 10^{-6}$ and $P_0 = 0.5368$. (a) Optimal control curve u^*. (b-d) Numbers of newly confirmed cases, hospitalized individuals, and the proportion of the population adopting protective behavior under optimal control, respectively. The black dashed lines indicate the equilibrium values of the corresponding variables for the uncontrolled system (6) (color figure online)

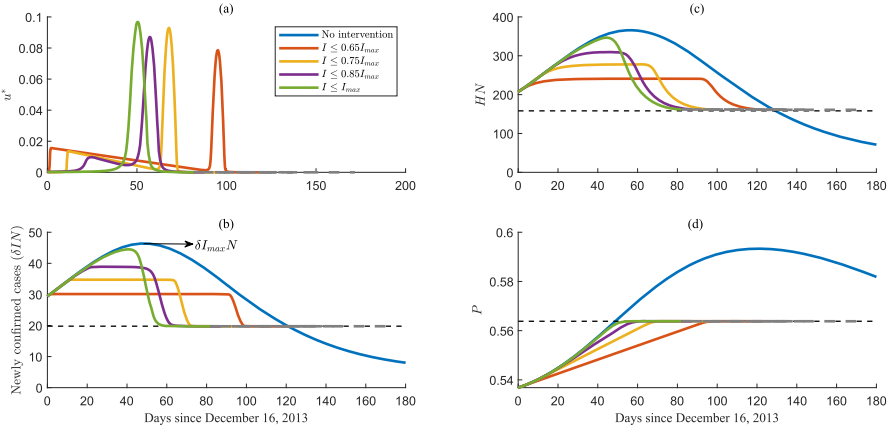


Fig. 9 Results of the single-control OCP under different maximum prevalence constraints. The initial values are $I_0 = 1.9153 \times 10^{-5}$, $H_0 = 5.6638 \times 10^{-6}$ and $P_0 = 0.5368$. (a) Optimal control curve u^*. (b-d) Numbers of newly confirmed cases, hospitalized individuals, and the proportion of the population adopting protective behavior under optimal control, respectively. The black dashed lines indicate the equilibrium values of the corresponding variables for the uncontrolled system (color figure online)

A single, short intervention then suffices to steer the system to a sustainable equilibrium without significant rebounds. However, if initiated beyond this window near the epidemic peak, the optimal strategy shifts to two intensive interventions requiring a substantially stronger effort, which increases control costs and triggers subsequent rebounds. Further delaying initiation until after the peak can render even two high intensity, concentrated interventions may fail to converge to the desired equilibrium before the next wave. In this scenario, infection and hospitalization peaks may exceed

those of the uncontrolled case, with more pronounced rebounds. These results confirm that achieving a low-level epidemic requires early surveillance, prompt initiation of optimization, and timely intervention within the "golden window" of rapid epidemic growth. Delayed action, by contrast, leads to strategic reversals, higher costs, and a loss of public trust.

Then, we explore how the public response rate affects the implementation of control strategies and the resulting epidemic outcomes. By setting December 16, 2013 as the optimization start date and varying the public response rate $\tilde{\varepsilon}$, the results, shown in Fig. 8, demonstrate that a faster public response leads to earlier interventions with milder intensity and shorter duration, thereby reducing dependence on external measures as adaptive public behavior plays a greater role. This also suppresses the infection peak more effectively and raises overall control efficiency. Thus, enhancing public response rates contributes to both cost reduction and epidemic control. This emphasizes the necessity for policies that integrate external measures with behavioral guidance to promote timely public adaptation.

Finally, we extend the unconstrained OCP (11) to explicitly incorporate state constraints and use numerical methods to investigate the effects of an upper bound on prevalence on optimal strategies and epidemic outcomes. Imposing this constraint is essential because prevalence often surpasses manageable thresholds before hospital pressure emerges, inevitably leading to overcrowding. The constraint enables early and proactive containment, curbing the infection peak, reducing severe cases, and preserving healthcare capacity. In the numerical experiments, we set the optimization start date to December 16, 2013, to steer the system toward the target equilibrium x^*. As shown in Fig. 9, stricter prevalence constraints significantly alter the optimal intervention in pattern, timing, and duration. Specifically, tighter bounds prolong the total intervention period and shift the strategy from a single concentrated to two phases. An earlier, stronger, and longer-lasting initial intervention contrasts with a delayed secondary response, which prolongs the plateau near the peak. Consequently, stricter constraints on peak require policymakers to mobilize resources earlier and cope with the managerial pressure of prolonged, multi-phase interventions.

5 Extension from Single to Dual Control: Application to COVID-19

For long-circulating infections like influenza, public risk perception is generally stable. In emerging outbreaks, however, authorities can not only lower the basic reproduction number via NPIs (control variable u_1), but also shape behavioral responses by adjusting risk signals through the control variable u_2 applied to the behavioral threshold I_c, subject to $I_c > u_2$. Our goal is to find the optimal strategy $\mathbf{u}^*(t) = [u_1^*,\ u_2^*]^\top$ that drives the system into a predefined neighborhood (10^{-6}) of the target equilibrium x^* in the shortest time T. Note that while the endemic equilibrium depends on $\mathbf{u}$, maintaining $\mathbf{u}(t) = (0,\ I_c - I^*)^\top$ for $t > T$ allows the disease to be sustained at the target prevalence by fixing a single behavioral threshold.

To limit intensity of NPIs, $u_{1\max}$ is set to 0.7 so as to cover intervention levels comparable to a full-city lockdown. Meanwhile, to prevent excessive protection costs arising from an overly low behavioral threshold, or delayed public response and

heightened hospital demand resulting from an overly high one, the adjustment of risk signals is constrained by $u_{2\min} \le u_2(t) \le u_{2\max}$. Specifically, for sustainability of the target equilibrium, we set $u_{2\min} = -(I_c - I^*)$ and $u_{2\max} = I_c - I^*$. Thus, we define the admissible control set $\overline{U} = [0, u_{1\max}] \times [u_{2\min}, u_{2\max}]$. The corresponding cost functional of OCP is formulated as follows:

$$\overline{J}(x(t), \mathbf{u}(t)) = hI(t) + \rho(1-h)u_1^2(t) + (1-\rho)(1-h)u_2^2(t),$$

where $h \in [0, 1]$ denotes the disease cost weight coefficient, and $\rho \in [0, 1]$ is the control cost allocation coefficient. We aim to determine the optimal control strategy by solving the following OCP:

$$\begin{aligned}
&\min_{\mathbf{u}(t)} \int_{t_0}^{T} \overline{J}(x(t), \mathbf{u}(t))\, dt + \Phi(x(T)), \\
\text{subject to} \quad & I' = (1-u_1(t))\, R_0\, [(1-P) + (1-a)\, P]\, (1-I-H)I - I, \\
& H' = \delta I - \tau H, \\
& P' = \tilde{\varepsilon} \left(\frac{I}{I_c - u_2(t)} - 1 \right) P\,(1-P)\,. \\
& x(t_0) = x_0, \\
& 0 \le u_1(t) \le u_{1\max},\ u_{2\min} \le u_2(t) \le u_{2\max}.
\end{aligned} \tag{12}$$

For the dual-control optimal control problem (12), the existence of an optimal control pair can be established analogously to Theorem 3. Therefore, we omit the proof here. The expression of the optimal control strategy is given in the following theorem.

Theorem 5 *For the OCP (12), there exist adjoint variables* $\overline{\boldsymbol{\lambda}}(t) = [\overline{\lambda}_1(t), \overline{\lambda}_2(t), \overline{\lambda}_3(t)]$, *and an optimal control* $\mathbf{u}^*(t)$, *which minimizes the cost functional over the admissible control set* $\overline{U}$, *and* $\mathbf{u}^*(t)$ *satisfies the following necessary condition:*

$$u_1^*(t) = \min \left\{ u_{1\max},\ \max \left\{ 0,\ \frac{\overline{\lambda}_1(t) R_0 (1 - aP(t))(1 - I(t) - H(t))I(t)}{2\rho(1-h)} \right\} \right\},$$

$$u_2^*(t) = \underset{u_2(t) \in [u_{2\min}, u_{2\max}]}{\arg\min} \left\{ (1-\rho)(1-h)u_2^2(t) + \overline{\lambda}_3(t)\tilde{\varepsilon} \frac{P(t)(1-P(t))I(t)}{I_c - u_2(t)} \right\},$$

and the adjoint variables satisfy the transversality condition associated with the terminal cost $\overline{\boldsymbol{\lambda}}(T) = \nabla_x \phi(x(T))$.

Detailed derivations of the adjoint functions $\overline{\lambda}_i$ $(i = 1, 2, 3)$ and the solution of the OCP (12) are given in Appendix E.

OCP (12) was applied to the first wave of the COVID-19 epidemic in Shanghai, 2022, with parameters estimated by fitting the model-simulated daily new infections to the daily reported positive cases from February 24 to April 21 (Fig. 10(a)).

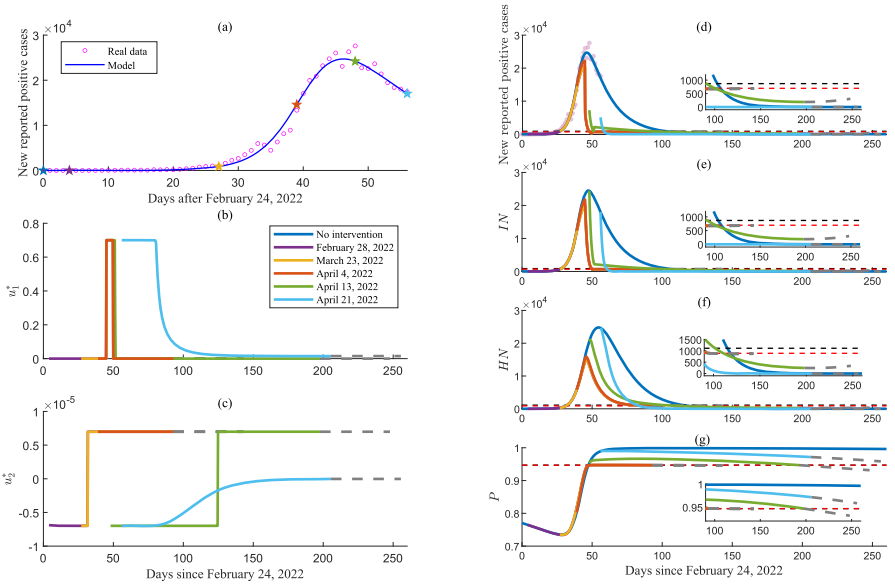


Fig. 10 (a) Fitting result for daily reported positive cases. (b-g) Results of the OCP (12) under different optimization starting times, with the endemic equilibrium point at a prevalence of $80\%I_c$ as the terminal target. Model parameters: $R_0 = 2.1727$, $N = 2489.4 \times 10^4$, $\tau = 0.125$, $a = 0.57$, $\tilde{\varepsilon} = 7.9112 \times 10^{-3}$, $I_c = 3.4936 \times 10^{-5}$, $\delta = 0.16$, $h = 0.6$, $\psi = 3 \times 10^8$, $\rho = 0.1$, $I_0 = 4.017 \times 10^{-8}$, $H_0 = 0$ and $P_0 = 0.7705$. (b-c) Optimal control curves for u_1 and u_2, respectively. (d-g) Numbers of newly infected individuals, total infections, hospitalized individuals, and the proportion of the population adopting protective behavior under optimal control, respectively. The black and red dashed lines represent the values of the corresponding variables at the endemic equilibrium of the uncontrolled system (6) and at the endemic equilibrium corresponding to a prevalence of $80\%I_c$, respectively (color figure online)

We first examine an OCP (12) that guides the system toward an endemic equilibrium with lower prevalence ($80\%I_c$), starting optimization at different times. The optimization starting time also significantly affects the optimal structure and switching timing of the two control variables. When the optimization is initiated during the rapid growth phase, both u_1 and u_2 show bang-bang structures (Fig. 10(b,c)). Although u_2 switches earlier than u_1, their start and end times are consistent across different initiation times, indicating a stable critical time window. Specifically, u_1 remains inactive in the early stage and applies only a short high-intensity control later, while u_2 first raises and then lowers the behavioral threshold, thereby guiding the system toward a long-term controllable steady state. If optimization starts near the epidemic peak, both controls remain bang-bang, but u_2 now switches later than u_1. High-intensity u_1 must be applied immediately to compensate for accumulated infections due to the delay. When initiated after the peak, the bang-bang structure is lost: u_1 remains at maximum before gradual relaxation, and u_2 increases monotonically, maintaining behavior above the baseline I_c. In this delayed scenario, early infection accumulation induces persistent protective behavior, and even under strong combined controls, behavioral inertia prevents the system from converging timely to the target equilibrium, as shown in Fig. 10(e,g). These findings highlight the importance of early identification of the

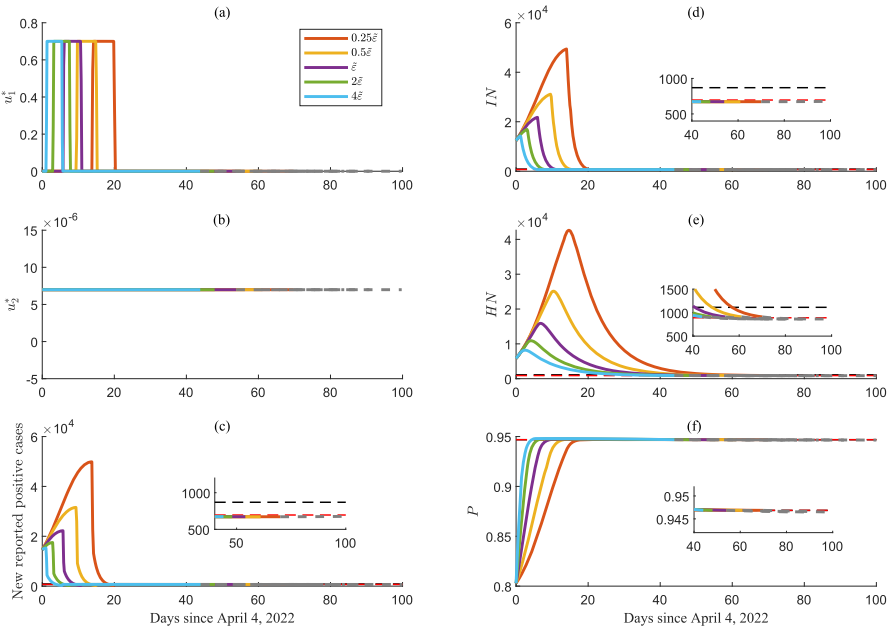


Fig. 11 Results of the OCP (12) under different behavioral response rates $\tilde{\varepsilon}$, with the endemic equilibrium point at a prevalence of 80%I_c as the terminal target. The initial values are $I_0 = 4.97804 \times 10^{-4}$, $H_0 = 2.3392 \times 10^{-4}$ and $P_0 = 0.8029$. (a-b) Optimal control curves for u_1 and u_2, respectively. (c-f) Numbers of newly infected individuals, total infections, hospitalized individuals, and the proportion of the population adopting protective behavior under optimal control, respectively. The black and red dashed lines represent the values of the corresponding variables at the endemic equilibrium of the uncontrolled system (6) and at the endemic equilibrium corresponding to a prevalence of 80%I_c, respectively (color figure online)

critical optimization window. Missing it allows epidemic inertia and behavioral delays to undermine control, delaying targets, raising costs, and reducing intervention efficiency. During the rapid growth phase, timely deployment of NPIs within this golden window is essential for cost-effective containment.

Next, taking April 4, 2022 as the optimization start date, we simulated optimal control paths under different public response rates $\tilde{\varepsilon}$. As shown in Fig. 11, the optimal control u_1 follows a bang-bang structure, switching rapidly to maximum intensity for a brief period. A faster public response rate leads to an earlier onset of u_1, which is implemented alongside a lower and stable behavioral threshold. This strategy drives a rapid rise in protective behavior, establishing early population-level shielding, more effectively suppressing the infection peak, and shortening the time to reach the target. These findings imply that increasing public response speed amplifies the effect of external interventions and enables more efficient, lower-cost epidemic control.

Then, we investigate the OCP(12) for three target prevalence levels (80%, 50%, 20% of the endemic prevalence I_c of system (6)), starting interventions on April 4. Fig. 12 shows that the optimal control u_1 exhibits a bang-bang control structure. Lower targets lead to earlier onset and longer high-intensity control. In contrast, the threshold adjustment u_2 remains constant, with its value increasing for lower targets, reflecting

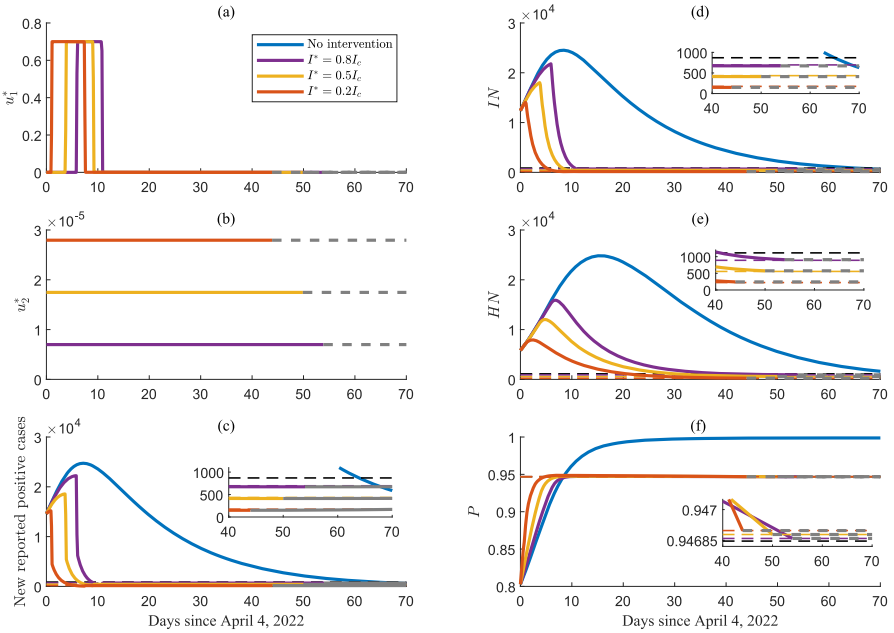


Fig. 12 Results of the OCP (12) with the endemic equilibria corresponding to different prevalence levels as terminal targets. The initial values are $I_0 = 4.97804 \times 10^{-4}$, $H_0 = 2.3392 \times 10^{-4}$ and $P_0 = 0.8029$. (a-b) Optimal control curves for u_1 and u_2, respectively. (c-f) Numbers of newly infected individuals, total infections, hospitalized individuals, and the proportion of the population adopting protective behavior under optimal control, respectively. The black dashed line represents the variable values at the equilibrium of the uncontrolled system (6). The colored dashed lines represent the variable values at the endemic equilibria corresponding to the prevalence indicated by their respective colors (color figure online)

a more easily activated behavioral threshold. This strategy reduces and advances both infection and hospitalization peaks, accelerating convergence. These results indicate that stringent containment requires immediately activating the highest level of NPIs while sustaining high vigilance against sporadic cases and consistent adherence to protection measures to establish community-wide defense and rapidly cut transmission chains.

Finally, we extend the analysis to include state constraints. This is motivated by reported data: during the outbreak, the peak number of hospitalized patients (25,010) remained below the city's designated hospital capacity of 30,000 beds (Shanghai Municipal Health Commission 2022). We therefore focus on an OCP that drives the system toward an endemic equilibrium with lower prevalence (i.e., $80\%I_c$) while limiting the maximum prevalence. Fig. 13 demonstrates that the maximum prevalence bound critically influences the intensity, timing and frequency of control. Under stricter constraints, the optimal strategy u_1 adopts an early, moderate intervention, which increases in strength as the constraint tightens. This is followed by a later bang-bang intervention that rapidly switches to and sustains the maximum allowed intensity to suppress prevalence quickly. In contrast, the behavioral threshold adjustment u_2 remains constant throughout. This control results in earlier plateaus for both new

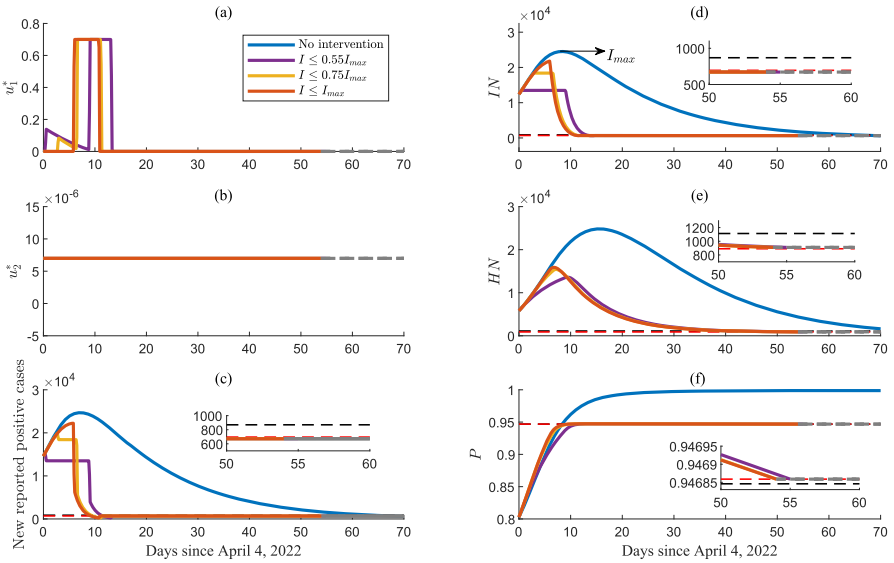


Fig. 13 Results of the dual-control OCP under different maximum prevalence constraints, with the endemic equilibrium point at a prevalence of 80%I_c as the terminal target. The initial values are $I_0 = 4.97804 \times 10^{-4}$, $H_0 = 2.3392 \times 10^{-4}$ and $P_0 = 0.8029$. (a-b) Optimal control curves for u_1 and u_2, respectively. (c-f) Numbers of newly infected individuals, total infections, hospitalized individuals, and the proportion of the population adopting protective behavior under optimal control, respectively. The black and red dashed lines represent the values of the corresponding variables at the endemic equilibrium of the uncontrolled system (6) and at the endemic equilibrium corresponding to a prevalence of 80%I_c, respectively (color figure online)

infections and the infection peak, alongside an extended plateau duration and a more gradual post-peak decline. These results underscore the indispensable role of early, moderate interventions under prevalence constraints. Compared to high-intensity centralized responses, a proactive and phased strategy achieves control objectives at a lower overall cost, thereby offering crucial theoretical support for dynamic resource allocation under limited capacity.

6 Conclusion and Discussion

This study emphasizes an overlooked reality of infectious disease control: the public actively adjusts protective behaviors based on real risk signals rather than passively facing infection. We propose a behavior-disease coupled model that uses healthcare resource accessibility as the risk signal and imitation dynamics for strategic selection, capturing their bidirectional feedback. Model analysis shows that the basic reproduction number R_0 without protective measures is the key threshold for epidemic eradication: the epidemic dies out when $R_0 < 1$; whereas an endemic equilibrium exists when $R_0 > 1$, determined by the behavioral decision threshold I_c. Moderately constrained healthcare resources act as effective risk signals that trigger proactive public protection and reduce prevalence.

We further design static and dynamic control strategies that balance prevalence and intervention costs. Inspired by (Parino et al. 2024), we adopt a terminal penalty to drive the system toward a desired equilibrium. Our key innovation is the introduction of a scenario-based control framework rooted in the stability of public risk perception. By distinguishing two typical scenarios based on epidemic characteristics, particularly whether public risk perception is long-term stable or short-term variable, we propose tailored control architectures accordingly. In the single-control case, we target R_0 rather than the willingness (or difficulty) of adopting protective behavior via the cost-risk trade-off (Parino et al. 2024), which is described by the behavioral threshold I_c. This is because R_0 can be directly reduced via non-pharmaceutical interventions (NPIs), such as gathering restrictions, while I_c reflects deep-seated public perceptions that are difficult to alter with short-term policy, especially for endemic, low-severity diseases. Thus, controlling R_0 is therefore more practical.

Analysis demonstrates that when only R_0 is controlled, the optimal fixed strategy depends on protection efficacy (a), behavioral threshold (I_c), and transmission risk weight (h). Given the relative stability of a, we focus on how h and I_c shape the optimal policy and system steady state. Numerical results indicate that higher h produces a non-decreasing optimal control that surges sharply within a critical range, demanding a different cost-containment strategy. The effect of I_c is more complex. In the colored transition zone of Fig. 5(c), a higher I_c weakens risk perception yet reduces optimal control, making the system sensitive to equilibrium shifts and necessitating precise external intervention to reach the target steady state. There also exists a region where optimal control naturally emerges from adaptive public behavior.

We apply the dynamic optimization framework to both influenza and COVID-19. Although both confer temporary immunity, viral mutations and antigenic shifts rapidly replenish the susceptible population, maintaining high susceptibility under long-term coexistence goals. Consequently, the SIHS model provides a rational simplification of this macroscopic dynamic. It focuses on optimal control under behavioral feedback while avoiding the complexity of increased system dimensionality, enabling direct strategy comparison between endemic influenza and emerging COVID-19 within a unified framework.

Analysis of influenza data from Shanxi Province reveals that risk-driven spontaneous behavioral evolution alone can generate recurrent multi-wave outbreaks. The staggered peaks of new cases and protective behavior indicate adaptive but delayed public responses. When only the reproduction number is controlled, we investigate how optimal strategy is influenced by the optimization start times, public response rates, and prevalence constraints. The results show that achieving low prevalence hinges on seizing the golden intervention window during the epidemic's rapid growth phase. Acting within this window allows a single brief control to steer the system toward a long-term manageable state without rebound. Delayed intervention, by contrast, alters the strategy, requiring more frequent, intensive controls and triggering inevitable resurgence, especially when the intervention initiated after the peak. A faster public response corresponds to an optimal control that starts earlier, requires lower peak intensity, and reaches the target in shorter time, along with more effective suppression of the infection peak. Tighter prevalence constraints extend the total intervention duration and shift the control pattern from a single, concentrated phase

to two phases: the first starts earlier, is stronger and lasts longer, while the second is delayed, prolonging the plateau at the epidemic peak.

For endemic influenza, public risk perception is relatively stable. In emerging outbreaks, however, authorities often need to both lower transmission via NPIs and adjust the behavioral threshold, prompting a dual-control strategy that targets both transmission and behavior.

Analysis of COVID-19 data from Shanghai confirms a critical time window during the rapid growth phase. Optimization initiated within this window yields bang-bang controls that switch simultaneously, with the behavioral threshold switching earlier. If delayed until near the peak, bang-bang controls remain but require immediate high-intensity NPIs to offset accumulated infections, and the switching order reverses. Starting after the peak shifts control to a non-bang-bang pattern, even with strong NPIs and behavioral threshold regulation, behavioral inertia prevents rapid convergence to the target. These findings underscore that identifying the golden intervention window early in the growth phase is key to reducing costs. Faster public response demands bang-bang NPIs that start earlier, coupled with a stable and lower behavioral threshold. This combination promotes swift population-wide protection and suppresses the epidemic effectively at lower cost and in a shorter time. Across different target steady states, more stringent containment targets correspond to the earlier initiation and more sustained implementation of NPIs, coupled with a lower threshold for triggering public protective behaviors. Under this strategy, both infection and hospitalization peaks occur earlier and are significantly reduced, and the system converges more rapidly to the target steady state. Under prevalence constraints, optimal NPIs adopt a two-stage strategy: an initial weak phase followed by a bang-bang phase, with tighter constraints corresponding to stronger initial control, while the behavioral-threshold regulation remains constant. This strategy leads to an earlier onset and a significantly prolonged epidemic plateau, with a slower decline rate after the peak.

Future research can extend the current model by integrating diverse interventions tailored to diseases with different transmission characteristics. For instance, introducing both pharmaceutical and non-pharmaceutical measures into the optimal control framework allows corresponding control variables to be added to the transmission and recovery terms. This enables systematic analysis of their synergies, trade-offs, and complementary effects, as well as optimized allocation under resource constraints. Such an extension would improve the model's adaptability to complex control scenarios and provide policymakers with more practical, flexible decision support. Furthermore, in the current model, the feasible domain of P is $[0, 1]$, implying that all individuals can switch between protective states. In reality, stubborn individuals who always or never adopt protection may restrict the feasible range to a subinterval $[p_1, p_2] \subset [0, 1]$. Recently, (Cai et al. 2025) introduced a behavioral equation of the form $(P - p_1)(p_2 - P)$, which implicitly confines behavioral imitation to the subpopulation with adjustable behaviors, effectively capturing the boundary characteristics of the protection proportion and providing a significant reference for phenomenological modeling. Inspired by the work of (Cai et al. 2025), in future research we plan to incorporate stubborn individuals into the model, for instance through heterogeneous classification, to enhance the model's explanatory power for real-world scenarios.

This study has several limitations that suggest directions for future work. First, it shares the common epidemiological limitation of reporting rate uncertainty. Underreporting associated with asymptomatic infections and reporting delays affects parameter estimation and optimal control strategies. Future work incorporating time-varying or unknown reporting rates would enhance model realism and control precision. Second, public risk perception involves more than healthcare availability, requiring the integration of multiple information sources and psychological factors. In addition, the linear risk assumption, while tractable, overlooks potential threshold or saturation effects, pointing to nonlinear functions as a direction for future work. Finally, the assumption of homogeneous mixing ignores population heterogeneity, which should be considered to improve the realism and accuracy of the model.

Appendix A Proof of Theorem 1

Proof: To study the stability of the equilibria, we compute the Jacobian matrix of system (6) at a general point, given by

$$J_x = \begin{bmatrix} R_0\,(1-aP)\,(1-2I-H)-1 & -R_0\,(1-aP)\,I & -aR_0\,(1-I-H)\,I \\ \delta & -\tau & 0 \\ \frac{\tilde{\varepsilon}p(1-P)}{I_c} & 0 & \tilde{\varepsilon}\left(\frac{I}{I_c}-1\right)(1-2P) \end{bmatrix}.$$

(1) The Jacobian matrix at DFE_1 is

$$J_{DFE_1} = \begin{bmatrix} R_0-1 & 0 & 0 \\ \delta & -\tau & 0 \\ 0 & 0 & -\tilde{\varepsilon} \end{bmatrix}.$$

It is straightforward to derive that the eigenvalues of J_{DFE_1} are R_0-1, $-\tau$ and $-\tilde{\varepsilon}$, respectively. Therefore, DFE_1 is stable if $R_0 < 1$.

(2) For EE_1, it is evident that when $R_0 > 1$, EE_1 exists. The Jacobian matrix at EE_1 is given by

$$J_{EE_1} = \begin{bmatrix} -\frac{\tau(R_0-1)}{\delta+\tau} & -\frac{\tau(R_0-1)}{\delta+\tau} & -\frac{a(R_0-1)\tau}{R_0(\delta+\tau)} \\ \delta & -\tau & 0 \\ 0 & 0 & \tilde{\varepsilon}\left(\frac{(R_0-1)\tau}{I_cR_0(\delta+\tau)}-1\right) \end{bmatrix}.$$

One of the eigenvalues of J_{EE_1} is $\lambda_1 = \tilde{\varepsilon}\left(\frac{\tau}{I_c(\delta+\tau)}\left(1-\frac{1}{R_0}\right)-1\right)$.

For the sub-matrix

$$A = \begin{bmatrix} -\frac{\tau(R_0-1)}{\delta+\tau} & -\frac{\tau(R_0-1)}{\delta+\tau} \\ \delta & -\tau \end{bmatrix}.$$

its trace is $\mathrm{trace}(A) = -\frac{\tau(R_0-1)}{\delta+\tau} - \tau < 0$, and its determinant is $\det(A) = \tau\,(R_0-1) > 0$. Consequently both eigenvalues of A have negative real parts, and

the stability of EE_1 is determined by λ_1. This implies that EE_1 is stable when $I_c > \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right)$.

(3) DFE_2 exists for all parameter regimes, and the Jacobian matrix at DFE_2 is

$$J_{DFE_2} = \begin{bmatrix} (1-a)\,R_0 - 1 & 0 & 0 \\ \delta & -\tau & 0 \\ 0 & 0 & \tilde{\varepsilon} \end{bmatrix}.$$

Given that J_{DFE_2} has an eigenvalue $\tilde{\varepsilon} > 0$, it implies that DFE_2 is an unstable saddle point.

(4) For EE_2, it is easy to see that EE_2 exists when $(1-a)\,R_0 > 1$. The Jacobian matrix at EE_2 is

$$J_{EE_2} = \begin{bmatrix} -\frac{\tau((1-a)R_0-1)}{\delta+\tau} & -\frac{\tau((1-a)R_0-1)}{\delta+\tau} & -\frac{a\tau((1-a)R_0-1)}{(1-a)^2R_0(\delta+\tau)} \\ \delta & -\tau & 0 \\ 0 & 0 & -\tilde{\varepsilon}\left(\frac{\tau((1-a)R_0-1)}{I_c(1-a)R_0(\delta+\tau)}-1\right) \end{bmatrix}.$$

It is evident that one eigenvalue of J_{EE_2} is $\lambda_1 = \tilde{\varepsilon}\left(\frac{\tau}{I_c(\delta+\tau)}\left(1-\frac{1}{(1-a)R_0}\right)-1\right)$.

Let

$$B = \begin{bmatrix} -\frac{\tau((1-a)R_0-1)}{\delta+\tau} & -\frac{\tau((1-a)R_0-1)}{\delta+\tau} \\ \delta & -\tau \end{bmatrix},$$

it trace and determinant satisfy $\text{trace}(B) = -\frac{\tau((1-a)R_0-1)}{\delta+\tau} - \tau < 0$, $\det(B) = \tau\left((1-a)\,R_0 - 1\right) > 0$, which implies that both eigenvalues of B are negative real parts. Hence, the stability of EE_2 is determined by λ_1, and EE_2 is stable when $I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{(1-a)R_0}\right)$.

(5) Next, we consider the equilibrium EE_3, its existence requires the simultaneous satisfaction of two conditions: $I_c + \frac{\delta I_c}{\tau} \in (0,1]$ and $\frac{R_0(\tau-\tau I_c-\delta I_c)-\tau}{aR_0(\tau-\tau I_c-\delta I_c)} \in (0,1)$. We examine each condition in turn.

(i). The first condition simplifies to: $I_c + \frac{\delta I_c}{\tau} \in (0,1] \Leftrightarrow I_c + \frac{\delta I_c}{\tau} \le 1 \Leftrightarrow \tau - \tau I_c - \delta I_c \ge 0$;

(ii). For the second condition, we note that the expression for P_5 is undefined if $\tau - \tau I_c - \delta I_c = 0$. Hence the strict inequality $\tau - \tau I_c - \delta I_c > 0$ must hold. Only then is the second fraction well-defined, and we can proceed to check the bound $(0,1)$. Specifically,

$$\frac{R_0\,(\tau-\tau I_c-\delta I_c)-\tau}{aR_0\,(\tau-\tau I_c-\delta I_c)} \in (0,1), \;\Leftrightarrow \begin{cases} \tau-\tau I_c-\delta I_c > 0, \\ R_0(\tau-\tau I_c-\delta I_c)-\tau > 0, \\ R_0\,(\tau-\tau I_c-\delta I_c)-\tau < aR_0\,(\tau-\tau I_c-\delta I_c)\,, \end{cases}$$

$$\Leftrightarrow \begin{cases} R_0 > 1, \\ (1-a)R_0 > 1, \\ \frac{\tau}{\delta+\tau}\left(1-\frac{1}{(1-a)\,R_0}\right) < I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right), \end{cases} \text{or} \begin{cases} R_0 > 1, \\ (1-a)R_0 \le 1, \\ I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right). \end{cases}$$

Therefore, when $R_0 > 1$, EE_3 exists if one of the following conditions is satisfied: (a) $(1-a)R_0 > 1$ and $\frac{\tau}{\delta+\tau}\left(1-\frac{1}{(1-a)R_0}\right) < I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right)$; or (b) $(1-a)R_0 \le 1$ and $I_c < \frac{\tau}{\delta+\tau}\left(1-\frac{1}{R_0}\right)$. When EE_3 exists, the Jacobian matrix of system (6) at EE_3 is as follows:

$$J_{EE_3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ \delta & -\tau & 0 \\ a_{31} & 0 & 0 \end{bmatrix},$$

where $a_{11} = a_{12} = -\frac{\tau I_c}{\tau - \tau I_c - \delta I_c}$, $a_{13} = -\frac{aR_0(\tau - \tau I_c - \delta I_c)I_c}{\tau}$ and $a_{31} = \frac{\tilde{\varepsilon} P_5(1-P_5)}{I_c}$. When EE_3 exists, it is necessarily true tha $\tau - \tau I_c - \delta I_c > 0$, which ensures that $a_{11} = a_{12} < 0, a_{13} < 0, a_{31} > 0$. Thus, the characteristic polynomial of matrix J_{EE_3} is given by

$$\lambda^3 + b_1\lambda^2 + b_2\lambda + b_3 = 0,$$

where $b_1 = -(a_{11} - \tau) > 0, b_2 = -\tau a_{11} - a_{13}a_{31} - \delta a_{12} > 0, b_3 = -\tau a_{13}a_{31} > 0$.

$$H_1 = |b_1| > 0,$$
$$H_2 = \begin{vmatrix} b_1 & b_3 \\ 1 & b_2 \end{vmatrix} = a_{11}a_{13}a_{31} + a_{11}a_{12}\delta - \tau\delta a_{12} + a_{11}^2\tau - a_{11}\tau^2 > 0,$$
$$H_3 = \begin{vmatrix} b_1 & b_3 & 0 \\ 1 & b_2 & 0 \\ 0 & b_1 & b_3 \end{vmatrix} = b_3 H_2 > 0.$$

By the Routh - Hurwitz criterion, it is determined that EE_3 is always stable when it exists.

Appendix B Proof of Theorem 2

Recall that the cost functional $J(u)$ is decomposed into three segments on $u \in [0, 1]$ (see Section 4.1 for details). The subsequent derivations focus on analyzing the monotonicity and local minima of $J(u)$ across this interval to determine the optimal control strategy.

Proof: We begin by computing the partial derivative of $J_3(u)$ with respect to the control variable u. Direct calculation gives $\frac{\partial J_3}{\partial u} = 2(1-h)u \ge 0$, with equality if and only if $u = 0$. This indicates that J_3 is strictly increasing on $[0, 1]$. The continuity of $J(u)$ guarantees that $u = 1 - C$ is never the optimal control.

Next, we analyze the monotonicity of J_1 and J_2. For the solution to $\frac{\partial J_1}{\partial u} = -\frac{h\tau}{(\delta+\tau)R_0(1-u)^2} + 2(1-h)u = 0$, which corresponds to the intersections of the curves defined by $A = \frac{h\tau}{2(1-h)(\delta+\tau)R_0} > 0$ and $f(u) = u(1-u)^2$. Given $f(0) = 0$, $f(1) = 0$ and $f'(u) = (1-u)(1-3u)$. Thus, $f(u)$ is monotonically increasing on $[0, \frac{1}{3}]$ and monotonically decreasing on $[\frac{1}{3}, 1]$, with its global maximum on $[0, 1]$ achieved at $f(\frac{1}{3}) = \frac{4}{27}$. Let

$$g(u) = u(1-u)^2 - A,$$

one readily verifies that the sign of $g(u)$ is equivalent to the sign of $\frac{\partial J_1}{\partial u}$, i.e.

$$g(u) > 0 \ (= 0, \ < 0) \quad \text{iff} \quad \frac{\partial J_1}{\partial u} > 0 \ (= 0, \ < 0).$$

Consequently, the number of intersections between the two curves (i.e., the solutions to $g(u)$=0) depends solely on the value of A, which can be categorized into the following two distinct cases: when $A \geq \frac{4}{27}$, the $g(u) \leq 0$ holds uniformly for all $u \in [0, 1]$, which implies that J_1 is monotonically decreasing on $[0, 1]$, whereas when $A < \frac{4}{27}$, the equation $g(u) = 0$ admits two distinct roots within $[0, 1]$, denoted by $u_1 \in (0, \frac{1}{3})$ and $u_2 \in (\frac{1}{3}, 1)$. In this case, $g(u) < 0$ for $u \in [0, u_1)$, indicating that J_1 is monotonically decreasing on this interval; $g(u) \geq 0$ for $u \in [u_1, u_2]$, where equality holds only at the endpoints, implying that J_1 is monotonically increasing on $[u_1, u_2]$; and $g(u) < 0$ for $u \in (u_2, 1]$, implying that J_1 is monotonically decreasing on $(u_2, 1]$.

For the solution to $\frac{\partial J_2}{\partial u} = -\frac{h\tau}{(\delta+\tau)(1-a)R_0(1-u)^2} + 2(1-h)u = 0$, which corresponds to the intersections of the two curves $B = \frac{h\tau}{2(1-h)(\delta+\tau)(1-a)R_0} > 0$ and $f(u) = u(1-u)^2$. Following a derivation analogous to the analysis of J_1 above, one yields that when $B \geq \frac{4}{27}$, J_2 is monotonically decreasing on $[0, 1]$, whereas when $B < \frac{4}{27}$, the two curves intersect twice within $[0, 1]$, with the intersection points denoted by $v_1 \in (0, \frac{1}{3})$ and $v_2 \in (\frac{1}{3}, 1)$. Thus, J_2 is monotonically decreasing on $[0, v_1)$; monotonically increasing on $[v_1, v_2]$; monotonically decreasing again on $(v_2, 1]$.

According to the definitions of A and B, it is clear that $A < B$ implies $u_1 < v_1 < v_2 < u_2$, which in turn allows the monotonicity of each segment in the cost functional to be classified into three distinct regimes. When $B > A \geq \frac{4}{27}$, both J_1 and J_2 are monotonically decreasing on $[0, 1]$. When $B \geq \frac{4}{27} > A$, J_1 exhibits a piecewise monotonic behavior: it is decreasing on $[0, u_1)$ and $(u_2, 1]$, while increasing on $[u_1, u_2]$. Meanwhile, J_2 is monotonically decreasing throughout $[0, 1]$. When $\frac{4}{27} > B > A$, both J_1 and J_2 exhibit piecewise monotonic behavior: J_1 is decreasing on $[0, u_1)$ and $(u_2, 1]$, while increasing on $[u_1, u_2]$. J_2 is decreasing on $[0, v_1)$ and $(v_2, 1]$, while increasing on $[v_1, v_2]$.

We will now explore three distinct scenarios based on the interval in which $u = 0$ belongs. For each scenario, we will further analyze the three cases corresponding to the aforementioned values of A and B.

I. $1 - C \leq 0$. In this situation, the entire interval $[0, u_u]$ is contained in Ω_1. When $B > A \geq \frac{4}{27}$, the function $J(u)$ is strictly decreasing on $[0, u_u]$, and consequently the global minimizer is attained at u_u.

If instead either $\frac{4}{27} > B > A$ or $B \geq \frac{4}{27} > A$, then $J_1(u)$ admits two critical points: a local minimum u_1 and a local maximum u_2. The global minimum of $J(u)$ depends on their placement relative to u_u. When $u_2 < u_u$, the function decreases on both $[0, u_1)$ and $(u_2, u_u]$, and increases on $[u_1, u_2]$; therefore, the global minimum is given by comparing $J(u_1)$ and $J(u_u)$. When the ordering is $u_1 < u_u < u_2$, the cost decreases up to u_1 and then increases, and hence u_1 yields the minimum. Finally, if $u_1 > u_u$, then $J(u)$ decreases throughout $[0, u_u]$, and the minimum again occurs at u_u.

II. $1 - D \leq 0 < 1 - C$. In this case, the feasible interval is divided such that $[0, 1-C] \subset \Omega_3$ and $(1-C, u_u] \subset \Omega_1$. Since $J(u)$ is strictly increasing on $[0, 1-C]$, the value $J(0)$ must be included among the candidates for the global minimum, to be compared with potential minimizers arising in Ω_1.

When $B > A \geq \frac{4}{27}$, the function $J(u)$ decreases on $[1-C, u_u]$. Hence the optimal value is obtained by comparing $J(0)$ with $J(u_u)$.

If either $\frac{4}{27} > B > A$ or $B \geq \frac{4}{27} > A$ holds, then the relative positions of u_1 and u_2 with respect to the boundary points $1 - C$ and u_u determine the monotonicity of $J(u)$ within Ω_1. If $u_2 < 1 - C$ or $u_1 > u_u$, then $J(u)$ is decreasing on $[1 - C, u_u]$, and the global minimum again reduces to comparing $J(0)$ and $J(u_u)$. When the ordering is $u_1 < 1 - C < u_2 < u_u$, the function $J(u)$ increases on $[1 - C, u_2]$ and decreases on $[u_2, u_u]$; in this situation, the global minimum follows from comparing $J(0)$ and $J(u_u)$. If $u_1 < 1-C$ and $u_u < u_2$, then $J(u)$ increases on the entire interval $[1 - C, u_u]$, which implies that $J(0)$ is the global minimum. When the critical points satisfy $1 - C < u_1 < u_2 < u_u$, the function decreases on $[1 - C, u_1)$, increases on $[u_1, u_2]$, and decreases again on $(u_2, u_u]$. Local minima in Ω_1 occur at u_1 or u_u, and the global minimum is obtained by comparing $J(0)$, $J(u_1)$ and $J(u_u)$. Finally, if $1 - C < u_1 < u_u < u_2$, the function $J(u)$ decreases on $[1 - C, u_1)$ and increases on $[u_1, u_u]$; in this case, the unique minimum inside Ω_1 occurs at u_1, and the global minimum is obtained by comparing $J(0)$ and $J(u_1)$.

III. $0 < 1-D$. In this case, the feasible interval splits into three parts: $[0, 1-D) \subset \Omega_2$, $[1 - D, 1 - C] = \Omega_3$ and $(1 - C, u_u] = \Omega_1$.

When $B > A \geq \frac{4}{27}$, $J(u)$ decreases on $[0, 1-D)$ and $(1-C, u_u]$, while it increases on $[1 - D, 1 - C]$. Hence the global minimum is determined by comparing $J(1 - D)$ and $J(u_u)$.

When $B \geq \frac{4}{27} > A$, $J(u)$ decreases on $[0, 1 - D)$ and increases on $[1 - D, 1 - C]$, so that $u = 1 - D$ becomes a candidate for the global minimizer, in addition to the minima possibly arising in Ω_1. The monotonicity within Ω_1 depends on the locations of u_1 and u_2, and the corresponding analysis is exactly the same as in **Case 2** for the regime $B \geq \frac{4}{27} > A$. Accordingly, the following result can be directly obtained: If either $u_2 < 1 - C$ or $u_1 > u_u$, the minimum follows from comparing $J(1 - D)$ and $J(u_u)$. When $u_1 < 1 - C < u_2 < u_u$, the minimum is again between $J(1 - D)$ and $J(u_u)$. If $u_1 < 1 - C$ and $u_2 > u_u$, the continuity of the cost functional ensures a unique global minimum is $J(1 - D)$. When $1 - C < u_1 < u_2 < u_u$, the local minima of $J_1(u)$ occur at u_1 or u_u, which must be compared together with $J(1 - D)$. If $1 - C < u_1 < u_u < u_2$, the minimum is obtained by comparing $J(1 - D)$ and $J(u_1)$.

When $\frac{4}{27} > B > A$, the monotonicity in $[0, 1 - D)$ is governed by the positions of v_1 and v_2, while that in $(1 - C, u_u]$ is governed by the locations of u_1 and u_2. If both $v_2 < 1-D$ and $u_2 < 1-C$, $J(u)$ decreases on $[0, v_1)$, $(v_2, 1-D)$ and $(1-C, u_u]$, and increases on the remaining intervals. The global minimum is obtained by comparing $J(v_1)$, $J(1-D)$ and $J(u_u)$. When $v_2 < 1 - D$ and $1 - C < u_2 < u_u$, $J(u)$ decreases on $[0, v_1)$, $(v_2, 1 - D)$ and $(u_2, u_u]$, while it increases on $[v_1, v_2]$ and $[1 - D, u_2]$. The global minimum is determined by comparing $J(v_1)$, $J(1-D)$ and $J(u_u)$. If $v_2 < 1-D$ and $u_2 > u_u$, the cost increases on $[0, v_1)$ and $(v_2, 1 - D)$, and decreases on $[v_1, v_2]$ and $[1 - D, u_u]$. Consequently, only $J(v_1)$ and $J(1 - D)$ remain as candidates for

the global minimum. If both $v_1 < 1 - D < v_2$ and $u_2 < 1 - C$, $J(u)$ decreases on $[0, v_1)$ and $(1 - C, u_u]$, and increases on $[v_1, 1 - C]$, the only relevant competitors are $J(v_1)$ and $J(u_u)$. If $v_1 < 1 - D < v_2$ and $1 - C < u_2 < u_u$, $J(u)$ decreases on $[0, v_1)$ and $(u_2, u_u]$, while increasing on $[v_1, u_2]$. The global minimum is determined by comparing $J(v_1)$ and $J(u_u)$. For $v_1 < 1 - D < v_2$ and $u_2 > u_u$, $J(u)$ decreases on $[0, v_1)$ and increases on $[v_1, u_u]$. The continuity of the cost functional ensures a unique global minimum is $J(v_1)$. If either $u_1 > u_u$ or $1 - D < v_1 < u_2 < 1 - C$, $J(u)$ decreases on $[0, 1 - D)$ and $(1 - C, u_u]$, and increases on $[1 - D, 1 - C]$. The global minimum is attained at either $1 - D$ or u_u. When $v_1 > 1 - D$ and $u_1 < 1 - C < u_2 < u_u$, $J(u)$ decreases on $[0, 1 - D)$ and $(u_2, u_u]$, while it increases on $[1 - D, u_2]$. The global minimum is occurs at either $1 - D$ or u_u. If $v_1 > 1 - D$, $u_1 < 1 - C$ and $u_2 > u_u$, then $J(u)$ decreases on $[0, 1 - D)$ and increases on $[1 - D, u_u]$, which makes $J(1 - D)$ the global minimizer. When $1 - C < u_1 < u_2 < u_u$, $J(u)$ decreases on $[0, 1 - D)$, $(1 - C, u_1)$ and $(u_2, u_u]$, while increasing on $[1 - D, 1 - C]$ and $[u_1, u_2]$. Thus, the global minimum must be selected from $J(1 - D)$, $J_h(u_1)$ and $J_h(u_u)$. Finally, if $1 - C < u_1 < u_u < u_2$, $J(u)$ decreases on $[0, 1 - D)$ and $(1 - C, u_1)$, and increases on $[1 - D, 1 - C]$ and $[u_1, u_u]$. In this case, the global minimum is determined by comparing $J(1 - D)$ and $J(u_1)$.

To conclude, the optimal strategy is drawn solely from the finite candidate set $\{0, v_1, u_1, 1 - D, u_u\}$, which completes the proof of Theorem 2.

Appendix C Proof of Theorem 3

Proof: The existence of an optimal control is established by verifying all conditions in Theorem 4.1 of (Fleming and Rishel 1975):

(1) The control set U and the corresponding state variable are nonempty.

(2) The control set U is closed and convex.

(3) The right-hand side $\vec{f}(t, x, u)$ of the state system is continuous and can be expressed as a linear function of u with coefficients depending on time and state. It is bounded by a linear function involving the state variable x and the control variable u.

(4) The integrand $J(x, u)$ of the objective functional is convex on U.

(5) There exist constants $a_1, a_2 > 0$, and $\beta^* > 1$ such that the integrand of the objective functional satisfies

$$J \geq \alpha_1 |u|^{\beta^*} - \alpha_2.$$

We now verify each condition in turn.

Verification of condition (1): The existence of an optimal control requires that solutions of the control system (10) remain bounded within finite time. This boundedness can be demonstrated by the fact that the supersolutions $\tilde{I}$, $\tilde{H}$, $\tilde{P}$ are bounded over a finite time interval.

$$\begin{cases} \tilde{I}'(t) = R_0 \tilde{I}, \\ \tilde{H}'(t) = \delta \tilde{I}, \\ \tilde{P}'(t) = \frac{\tilde{\varepsilon}}{I_c} \tilde{I}. \end{cases} \tag{13}$$

The system (13) is a linear system with bounded coefficients over finite time. As established in (Burden et al. 2004), the supersolutions $\tilde{I}$, $\tilde{H}$, $\tilde{P}$ are uniformly bounded.

Consequently, for every admissible control function in the set U, the solutions of the system remain bounded within a finite time interval. Furthermore, the right-hand side functions of system (10) are continuous and satisfy the Lipschitz condition with respect to the state variables. Therefore, by Theorem 9.2.1 in (Lukes 1982), the solutions of system (10) exist, which verifies condition (1).

Verification of condition (2): Following the definition of the set U, it is clear that U is closed and convex.

Verification of condition (3): The right-hand side of system (10) is continuous and linear in the control variable u, and can be written in the form:

$$\vec{f}(t,x,u) = \begin{pmatrix} R_0\,(1-aP)\,(1-I-H)I - I \\ \delta I - \tau H \\ \tilde{\varepsilon}\left(\frac{I}{I_c}-1\right)P\,(1-P) \end{pmatrix} + \begin{pmatrix} -R_0\,(1-aP)\,(1-I-H)I \\ 0 \\ 0 \end{pmatrix} u.$$

The boundedness of solutions yields

$$\left|\vec{f}(t,x,u)\right| \le \left|\begin{pmatrix} \tilde{I}' \\ \tilde{H}' \\ \tilde{P}' \end{pmatrix}\right| = \left|\begin{pmatrix} R_0 & 0 & 0 \\ \delta & 0 & 0 \\ \frac{\tilde{\varepsilon}}{I_c} & 0 & 0 \end{pmatrix}\begin{pmatrix} \tilde{I}' \\ \tilde{H}' \\ \tilde{P}' \end{pmatrix}\right| \le C_1(|x|+|u|+1),$$

where C_1 is a positive constant that depends on the coefficients of system (10).

Verification of condition (4): It is evident that the integrand $J(x,u)$ of the cost functional is convex with respect to the control variable u.

Verification of condition (5): There exist constants $\alpha_1 = 1-h > 0$, $\alpha_2 > 0$ and $\beta^* = 2$ such that the integrand of the cost functional satisfies

$$J(x,u) = hI + (1-h)u^2 \ge \alpha_1|u|^{\beta^*} - \alpha_2.$$

Therefore, there exists an optimal control pair that minimizes the cost functional.

Appendix D Proof of Theorem 4

Proof: To derive the expression of optimal control strategy, we begin by defining the Hamiltonian function as follows

$$\begin{aligned} \mathcal{H}(t,x,u,\boldsymbol{\lambda}) &= hI + (1-h)u^2 + \lambda_1 I' + \lambda_2 H' + \lambda_3 P' \\ &= hI + (1-h)u^2 + \lambda_1\left\{(1-u)R_0\big[(1-P)+(1-a)P\big](1-I-H)I - I\right\} \\ &\quad + \lambda_2(\delta I - \tau H) + \lambda_3\tilde{\varepsilon}\left(\frac{I}{I_c}-1\right)P(1-P), \end{aligned}$$

where λ_i $(i = 1, 2, 3)$ represent the adjoint variables associated with the state variables I, H, P respectively, and the corresponding adjoint equations are

$$\begin{aligned}
\lambda_1^{'} &= -\frac{\partial \mathcal{H}}{\partial I} = -h - \lambda_1(1-u)R_0(1-aP)(1-2I-H) \\
&\quad + \lambda_1 - \lambda_2\delta - \lambda_3\frac{\tilde{\varepsilon}}{I_c}P(1-P), \\
\lambda_2^{'} &= -\frac{\partial \mathcal{H}}{\partial H} = \lambda_1(1-u)R_0(1-aP)I + \lambda_2\tau, \\
\lambda_3^{'} &= -\frac{\partial \mathcal{H}}{\partial P} = a\lambda_1(1-u)R_0(1-I-H)I - \lambda_3\tilde{\varepsilon}\left(\frac{1}{I_c} - 1\right)(1-2P),
\end{aligned}$$

with the transversality condition $\boldsymbol{\lambda}(T) = \nabla_x \phi(x(T))$.

For the control variable u, the necessary optimality condition requires that the partial derivative of the Hamiltonian function with respect to u vanishes at the optimal control u^*. The partial derivative of $\mathcal{H}$ is given by

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial u} &= 2(1-h)u + \lambda_1\frac{\partial I'}{\partial u} + \lambda_2\frac{\partial H'}{\partial u} + \lambda_3\frac{\partial P'}{\partial u} \\
&= 2(1-h)u - \lambda_1 R_0(1-aP)(1-I-H)I.
\end{aligned}$$

Setting $\left.\frac{\partial \mathcal{H}}{\partial u}\right|_{u=u^*} = 0$ yields the optimality condition

$$\hat{u} = \frac{\lambda_1 R_0(1-aP)(1-I-H)I}{2(1-h)}.$$

To ensure the control variable remains feasible over the admissible set U, we have

$$u^*(t) = \min\left\{u_{\max}, \max\left\{0, \hat{u}(t)\right\}\right\}.$$

Appendix E Proof of Theorem 5

Proof: The Hamiltonian function associated with the OCP (12) is given by

$$\begin{aligned}
\overline{\mathcal{H}}(t, x, \mathbf{u}, \overline{\boldsymbol{\lambda}}) &= hI + \rho(1-h)u_1^2 + (1-\rho)(1-h)u_2^2 + \overline{\lambda}_1 I' + \overline{\lambda}_2 H' + \overline{\lambda}_3 P' \\
&= hI + \rho(1-h)u_1^2 + (1-\rho)(1-h)u_2^2 \\
&\quad + \overline{\lambda}_1[(1-u_1)R_0(1-aP)(1-I-H)I - I] \\
&\quad + \overline{\lambda}_2(\delta I - \tau H) + \overline{\lambda}_3\tilde{\varepsilon}\left(\frac{I}{I_c - u_2} - 1\right)P(1-P).
\end{aligned}$$

The adjoint variables $\overline{\lambda}_i$ $(i = 1, 2, 3)$ satisfy

$$\begin{aligned}
\overline{\lambda}_1^{'} &= -h - \overline{\lambda}_1(1-u_1)R_0(1-aP)(1-2I-H) + \overline{\lambda}_1 - \overline{\lambda}_2\delta - \overline{\lambda}_3\frac{\tilde{\varepsilon}}{I_c-u_2}P(1-P),\\
\overline{\lambda}_2^{'} &= \overline{\lambda}_1(1-u_1)R_0(1-aP)I + \overline{\lambda}_2\tau,\\
\overline{\lambda}_3^{'} &= a\overline{\lambda}_1(1-u_1)R_0(1-I-H)I - \overline{\lambda}_3\tilde{\varepsilon}\left(\frac{I}{I_c-u_2}-1\right)(1-2P),
\end{aligned}$$

with the transversality condition $\overline{\boldsymbol{\lambda}}(T) = \nabla_x\phi(x(T))$.

According to the PMP, the optimal control satisfies $\partial\overline{\mathcal{H}}/\partial u_i = 0$ for $i = 1, 2$, which gives

$$\begin{aligned}
\frac{\partial\overline{\mathcal{H}}}{\partial u_1} &= 2\rho(1-h)u_1 - \overline{\lambda}_1R_0(1-aP)(1-I-H)I,\\
\frac{\partial\overline{\mathcal{H}}}{\partial u_2} &= 2(1-\rho)(1-h)u_2 + \overline{\lambda}_3\tilde{\varepsilon}P(1-P)\frac{I}{(I_c-u_2)^2}.
\end{aligned}$$

Hence, one can derive the optimality conditions as follows

$$\hat{u}_1 = \frac{\overline{\lambda}_1R_0(1-aP)(1-I-H)I}{2\rho(1-h)},$$

$$2(1-\rho)(1-h)\hat{u}_2 + \overline{\lambda}_3\tilde{\varepsilon}P(1-P)\frac{I}{(I_c-\hat{u}_2)^2} = 0.$$

Taking the control constraints into account, the optimal control pair is

$$u_1^*(t) = \min\left\{u_{1\max},\ \max\left\{0, \hat{u}_1(t)\right\}\right\},$$

$$u_2^*(t) = \underset{u_2(t)\in[u_{2\min},u_{2\max}]}{\arg\min}\left\{(1-\rho)(1-h)u_2^2(t) + \overline{\lambda}_3(t)\tilde{\varepsilon}\frac{P(t)(1-P(t))I(t)}{I_c-u_2(t)}\right\},$$

where u_2^* is characterized implicitly by the minimization of $\overline{\mathcal{H}}$ with respect to u_2 over its admissible set.

Acknowledgements This research was partially supported by the National Key R&D Program of China (2025YFA1016502), the National Natural Science Foundation of China (92470107), and the Natural Science Foundation of Shanxi Province (202303021211003). Additionally, we are grateful to the editors and the anonymous referees for their careful reading and helpful comments which led to an improvement of our manuscript.

Data Availability The influenza data analyzed in this study were obtained from the Shanxi Provincial Center for Disease Control and Prevention (CDC) and available on request from the corresponding author, LiLi Liu. The COVID-19 data are obtained from the official reports of the Shanghai Municipal Health Commission, see https://wsjkw.sh.gov.cn/.

Declarations

Conflict of interest No potential conflict of interest was reported by the author(s).

References

Bauch CT (2005) Imitation dynamics predict vaccinating behaviour. Proc Biol Sci 272(1573):1669–1675. https://doi.org/10.1098/rspb.2005.3153

Bauch CT, Bhattacharyya S (2012) Evolutionary game theory and social learning can determine how vaccine scares unfold. PLoS Comput Biol 8(4):e1002452. https://doi.org/10.1371/journal.pcbi.1002452

Burden T, Ernstberger J, Fister KR (2004) Optimal control applied to immunotherapy. Discret Contin Dyn Syst B 4(1):135–146. https://doi.org/10.3934/dcdsb.2004.4.135

Cai CH, Huang JC, Lou YJ et al (2025) Coupled contagion dynamics of behavior and disease: A mechanistic learning model approach. J Data Dyn Syst 1:15–35. https://doi.org/10.1142/S3029286725500020

Capasso V, Serio G (1978) A generalization of the Kermack-McKendrick deterministic epidemic model. Math Biosci 42(1–2):43–61. https://doi.org/10.1016/0025-5564(78)90006-8

Chang SL, Piraveenan M, Pattison P et al (2020) Game theoretic modelling of infectious disease dynamics and intervention methods: a review. J Bio Dyn 14(1):1–33. https://doi.org/10.1080/17513758.2020.1720322

de Almeida CFCBR, da Silva ML, Junior WSF et al (2025) Can socioeconomic factors and the availability of medicinal plant resources influence people's perception of risk in relation to diseases? J Ethnobiol Ethnomed 21(35):1–18. https://doi.org/10.1186/s13002-025-00784-2

Fleming WH, Rishel RW (1975) Deterministic and Stochastic Optimal Control. Springer, New York

Geoffard PY, Philipson T (1996) Rational epidemics and their public control. Int Econ Rev 37(3):603–624. https://doi.org/10.2307/2527443

Gozzi N, Perra N, Vespignani A (2025) Comparative evaluation of behavioralepidemic models using COVID-19 data. PNAS 122(24):1–10. https://doi.org/10.1073/pnas.2421993122

Granell C, Gómez S, Arenas A (2013) Dynamical interplay between awareness and epidemic spreading in multiplex networks. Phys Rev Lett 111(12):128701. https://doi.org/10.1103/PhysRevLett.111.128701

Granell C, Gómez S, Arenas A (2014) Competing spreading processes on multiplex networks: Awareness and epidemics. Phys Rev E 90:012808. https://doi.org/10.1103/PhysRevE.90.012808

Hoertel N, Blachier M, Blanco C et al (2020) A stochastic agent-based model of the SARS-CoV-2 epidemic in France. Nat Med 26:1417–1421. https://doi.org/10.1038/s41591-020-1001-6

Huang M, Chen JQ (2024) Analysis of key influencing factors of public risk perception of public health emergencies in rural-urban fringe based on ISM and AHP. Saf Environ Eng 31(3):47–53

Huang YH, Zhu QY (2022) Game-theoretic frameworks for epidemic spreading and human decision-making: A review. Dyn Games Appl 12:7–48. https://doi.org/10.1007/s13235-022-00428-0

Javanian M, Barary M, Ghebrehewet S et al (2021) A brief review of influenza virus infection. J Med Virol 93(8):4638–4646. https://doi.org/10.1002/jmv.26990

Lau JTF, Yang XL, Tsui HY et al (2004) SARS related preventive and risk behaviours practised by Hong Kong-mainland China cross border travellers during the outbreak of the SARS epidemic in Hong Kong. J Epidemiol Community Health 58(12):988–996. https://doi.org/10.1136/jech.2003.017483

Li TJ, Xiao YN, Heffernan J (2024) Linking spontaneous behavioral changes to disease transmission dynamics: behavior change includes periodic oscillation. Bull Math Biol 86(73):1–55. https://doi.org/10.1007/s11538-024-01298-w

Liang MM, Gao L, Cheng C et al (2020) Efficacy of face mask in preventing respiratory virus transmission: A systematic review and meta-analysis. Travel Med Infect Dis 36:101751. https://doi.org/10.1016/j.tmaid.2020.101751

Lukes DL (1982) Differential Equations: Classical to Controlled. Academic Press, New York

Nunner H, Buskens V, Kretzschmar M (2021) A model for the co-evolution of dynamic social networks and infectious disease dynamics. Comput Soc Netw 8(19):1–33. https://doi.org/10.1186/s40649-021-00098-9

Parino F, Zino L, Rizzo A (2024) Optimal control of endemic epidemic diseases with behavioral response. IEEE Open J Control Syst 3:483–496. https://doi.org/10.1109/OJCSYS.2024.3488567

Perra N, Balcan D, GonÇalves B et al (2011) Towards a characterization of behavior-disease models. PLoS ONE 6(8):e23084. https://doi.org/10.1371/journal.pone.0023084

Pontryagin LS (1987) Mathematical Theory of Optimal Processes. CRC Press, London

Proverbio D, Tessarin R, Giordano G (2025) Data informed epidemiological-behavioural modelling. In: Marmo F, Cuomo S, Cutolo A (eds) Computational Mechanics and Applied Mathematics: Perspectives from Young Scholars. Springer, Cham, pp 279–289, https://doi.org/10.1007/978-3-031-76591-9_26

Putri WCWS, Muscatello DJ, Stockwell MS et al (2018) Economic burden of seasonal influenza in the United States. Vaccine 36(27):3960–3966. https://doi.org/10.1016/j.vaccine.2018.05.057
Sandholm WH (2010) Population Games and Evolutionary Dynamics. The MIT Press, Cambridge, MA
Seierstad A, Sydsæter K (1987) Optimal Control Theory with Economic Applications. North Holland, Amsterdam
Shanghai Municipal Health Commission (2022) 2022 shanghai municipal health status report. https://wsjkw.sh.gov.cn/2022wsjkzkbg/. Accessed 29 May 2025
Shanxi Provincial Bureau of Statistics (2019) Main data bulletin of sampling survey on population change in Shanxi Province. https://tjj.shanxi.gov.cn/tjsj/tjgb/qtpcgb. Accessed 26 May 2025
Silva PCL, Batista PV, Lima HS et al (2020) COVID-ABS: An agent-based model of COVID-19 epidemic to simulate health and economic effects of social distancing interventions. Chaos Solitons & Fractals 139:110088. https://doi.org/10.1016/j.chaos.2020.110088
Tang B, Zhou WK, Wang X (2022) Controlling multiple COVID-19 epidemic waves: An insight from a multi-scale model linking the behaviour change dynamics to the disease transmission dynamics. Bull Math Biol 84(106): https://doi.org/10.1007/s11538-022-01061-z
Veltri GA, Steinert JI, Sternberg H et al (2024) Assessing the perceived effect of non-pharmaceutical interventions on SARS-CoV-2 transmission risk: an experimental study in Europe. Sci Rep 14(4857):1–7. https://doi.org/10.1038/s41598-024-55447-1
WHO (2025) Influenza (seasonal). https://www.who.int/news-room/fact-sheets/detail/influenza-(seasonal). Accessed 26 May 2025
Xiao YN, Li Q, Zhou WK et al (2021) Analysis on dynamical mechanism of multi outbreaks of COVID-19. Chin J Epidemiol 42(6):966–976. https://doi.org/10.3760/cma.j.cn112338-20210219-00123
Zhang Y, Pan Y, Yang P et al (2015) Clinical characteristics of severe acute respiratory infection and risk factors associated with adverse outcomes in Beijing, 2014–2015 influenza season. Dis Surveill 30(12):1028–1032. https://doi.org/10.3784/j.issn.1003-9961.2015.12.010
Zhao S, Bauch CT, He DH (2018) Strategic decision making about travel during disease outbreaks: a game theoretical approach. J R Soc Interface 15(146):20180515. https://doi.org/10.1098/rsif.2018.0515
Zhao S, Stone L, Gao DZ et al. (2020) Imitation dynamics in the mitigation of the novel coronavirus disease (COVID-19) outbreak in wuhan, china from 2019 to 2020. Ann Transl Med 8(7):448. https://doi.org/10.21037/atm.2020.03.168

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